Learned Regularization for Robust Image Reconstruction:

A Comparative Study of Plug-and-Play Priors under Noise, Undersampling, Model Mismatch, and Distribution Shift

Mohammed Abo Oraig
Independent research
mohammed-abo-oraig@etud.univ-tln.fr

Abstract

We study sparse-view computed tomography as a linear inverse problem and compare classical variational regularization against a learned image prior used inside a Plug-and-Play proximal-gradient (PnP-PGD) scheme, benchmarking it against filtered back-projection, Tikhonov and total-variation (TV) regularization. We use disjoint train/validation/test splits, per-condition hyperparameter tuning, paired statistical tests, measurements simulated on a finer grid to avoid the inverse crime, and three independently trained denoisers per configuration. The learned prior improves PSNR by up to 4 dB over TV at moderate-to-high noise ($\sigma \geq 0.05$) and for 5–90 projection angles; in the near noise-free regime TV remains ahead, by an amount (1–6 dB) that depends on the forward-model mismatch of the simulated data. Reconstruction quality peaks after a handful of iterations and then degrades while the data residual rises. The advantage narrows under forward-model mismatch and reverses under distribution shift: a prior trained on MNIST loses to TV by 3–4 dB on Fashion-MNIST and on a Shepp–Logan phantom, whereas a prior trained on Fashion-MNIST beats TV on Fashion-MNIST by 2.2 dB, so the failure is one of prior mismatch rather than of the PnP scheme itself. An error decomposition along the singular vectors of the forward operator shows that, under mismatch, the reconstruction places up to three times the true energy in the weakly determined subspace. Finally, lesion-insertion experiments show that even in distribution the PSNR-tuned PnP reconstruction detects small inserted lesions less reliably than TV (paired AUC 0.69 vs. 0.91 for a 0.75 px lesion), attenuating them rather than inventing structure, whereas under prior mismatch it creates lesion-like blobs up to an order of magnitude more often than TV. Re-tuning towards lesion fidelity recovers TV-level detection only at the price of its PSNR advantage or of false structure. Finally we move to clinical-size data: 362×362 slices of LoDoPaB-CT with its Poisson photon model, patient-disjoint training, validation and test sets, 30, 60 and 120 of the 1000 projection angles, and one, one quarter and one tenth of the native dose. A DnCNN prior trained on 632 patients does not outperform a well-tuned TV in PSNR (differences between -0.93 and $+0.33$ dB over nine conditions) and has a lower SSIM in seven of them, but it detects small inserted lesions better than TV in all nine (paired AUC $+0.03$ to $+0.16$, significant in six), whereas TV tuned for PSNR detects them worse than unregularised filtered back-projection in all nine (significant in seven) despite a PSNR gain of 0.8–2.3 dB. On non-anatomical phantoms the detection advantage is retained in one of six conditions, the PSNR and SSIM deficit against TV is consistent, and the number of false lesion-like blobs relative to TV changes sign with the number of angles. We argue that task-based and robustness evaluation should be standard components of learned-prior reconstruction studies. We then replace the explicit forward matrix by a matrix-free operator with an exactly matched adjoint and the Gaussian noise by a Beer–Lambert Poisson model with 10–1000 photons per ray; the ranking of the methods, the reversal under distribution shift and the lesion results are unchanged, so they do not depend on a dense operator or on additive Gaussian noise.

1 Introduction

Image reconstruction from indirect, noisy, and incomplete measurements is a classical ill-posed inverse problem, arising in computed tomography (CT), magnetic resonance imaging, and many other imaging modalities. Given a forward operator A and noisy measurements

$$y = Ax + n, \tag{1}$$

the goal is to recover x from y when A is ill-conditioned or rank-deficient, as is the case for sparse-view CT with few projection angles. Classical approaches address ill-posedness through explicit regularization, solving

$$\hat{x} = \arg\min_x \ \tfrac{1}{2}\|Ax - y\|_2^2 + \lambda R(x), \tag{2}$$

with R a hand-crafted prior such as Tikhonov (L2) or total variation (TV) penalties. In recent years, learned image priors – most notably through Plug-and-Play (PnP) methods and deep unrolling – have been proposed to replace or augment hand-crafted regularizers with a denoising neural network, exploiting the empirical observation that image denoisers implicitly encode rich natural-image statistics [12].

While learned priors frequently outperform classical regularization on in-distribution benchmarks, a growing body of work has raised concerns about their *robustness*: sensitivity to distribution shift, lack of convergence guarantees, and instabilities under small perturbations of the input or forward model [1]. Many project-level or introductory studies of learned reconstruction report only in-distribution accuracy, leaving these robustness questions unaddressed.

This paper documents a controlled, small-scale study designed explicitly to probe these questions. We deliberately restrict scope to a simple, cheaply reproducible setting – sparse-view tomography of 28×28 MNIST-derived images – in order to run a thorough grid of experiments with proper train/validation/test protocol rather than a single headline comparison. Our contributions are:

- A from-scratch implementation of a sparse-view Radon operator with exact adjoint, classical regularization baselines (FBP, Tikhonov, TV-ADMM), and a PnP-PGD solver using a noise-conditioned DnCNN prior.
- A systematic comparison of PSNR/SSIM accuracy across measurement noise levels and numbers of projection angles, with validation-tuned hyperparameters and paired significance testing.
- A convergence analysis explaining why PnP-PGD selects very few iterations under validation tuning, showing a peak-then-decay pattern in reconstruction accuracy that is not visible from residual curves alone.
- Robustness studies – forward-model mismatch (angle offset and jitter) and train-test distribution shift (Fashion-MNIST, EMNIST, and a CT phantom) – run with hyperparameters frozen at their nominal values, exposing failure modes not visible under in-distribution evaluation alone.
- Data simulated on a finer grid than the reconstruction grid, to remove the inverse crime, with a sensitivity analysis over how the finer-grid object is constructed (Section 4.5).
- An error decomposition along the singular vectors of the forward operator, separating the weakly determined subspace from its orthogonal complement, with an "injection" and an "alignment" statistic that quantify prior-driven content (Section 4.6).

- Controls for the learned prior itself: three denoiser seeds, a wider training noise range, and a matched-prior control trained on the shifted distribution (Section 4.7).
- A task-based evaluation by lesion insertion, with an operating-point study asking whether the detectability loss can be removed by re-tuning (Sections 4.8 and 4.9).
- A matrix-free implementation of the projector and its exact adjoint, validated against the dense operator, and a Poisson (low-dose) noise study, showing that the conclusions above carry over to the settings required for clinical-size data (Sections 4.10–4.12).
- A patient-disjoint study on LoDoPaB-CT (362×362, exact Poisson dose thinning, crime-free simulation validated against the real observations to 0.15 dB) of three questions: whether the learned prior keeps its advantage over TV when the image distribution changes, whether it creates lesion-like structure that is not in the object, and whether the conclusions hold over a grid of three doses and three angle counts (Sections 4.13–4.18). The study reports negative and mixed results, a superseded early comparison, and the analysis steps taken after seeing results (Section 3.16).

2 Related Work

Variational regularization. Tikhonov regularization and total variation [10] remain standard baselines for ill-posed inverse problems due to their convexity, well-understood convergence properties, and interpretability. TV in particular is a strong prior for piecewise-constant images, which makes it a competitive baseline for digit-like data such as MNIST.

Plug-and-Play priors. Venkatakrishnan et al. [12] introduced the idea of replacing the proximal operator of a variational regularizer with an off-the-shelf denoiser inside an ADMM or proximal-gradient iteration. Subsequent work analyzed convergence conditions for PnP under assumptions on the denoiser (e.g. non-expansiveness or bounded Lipschitz constant) [5, 11]; without such constraints, plain CNN denoisers offer no convergence guarantee, a point directly relevant to the convergence behavior we observe in Section 4.

Learned denoisers. DnCNN [14] demonstrated that residual CNNs trained for Gaussian denoising across a range of noise levels generalize well as plug-in priors. We adopt this noise-conditioning strategy so that a single trained network can serve across the PnP iteration trajectory, where the effective denoising strength is a tunable hyperparameter.

Instabilities and robustness. Antun et al. [1] showed that learned reconstruction methods for CT and MRI can be unstable to small perturbations and structural changes not present in training data, in some cases more so than classical methods. Related work on distribution shift in learned inverse problem solvers [6] similarly finds that accuracy gains from learned priors can shrink or reverse under shift. Our distribution shift and forward-model mismatch experiments are a small-scale, project-level instantiation of this broader concern, and our findings (Section 5) are consistent with it.

Hallucinations, the inverse crime and task-based assessment. Bhadra et al. [3] define a hallucination in tomographic reconstruction as image content that is not determined by the measurement data, i.e. content that lives in the null space of the imaging operator and is supplied by

the prior; we adopt this view and extend it to the weakly determined subspace of a finite-precision operator (Section 3.6). Generating data with the same discrete model that is later inverted is known as the inverse crime [8, 13] and tends to overstate the accuracy of methods that exploit the exact model; we remove it in Section 3.5. Finally, we follow the task-based view of image quality [2]: fidelity metrics such as PSNR need not predict whether a clinically relevant feature survives reconstruction, so we also measure detection of inserted lesions.

3 Methodology

3.1 Forward operator

We construct a sparse-view Radon transform $A \in \mathbb{R}^{m \times n}$ as an explicit dense matrix by applying the discrete Radon transform to each canonical basis image and stacking the resulting sinogram columns. This gives an exact adjoint $A^\top$ (rather than an approximate one, as is common with implicit/fast Radon implementations), which is important for the PnP gradient step, and allows the reconstruction operator to be perturbed independently of the data-generating operator for the mismatch experiments in Section 4. Images are 28×28; the number of projection angles and, where relevant, an angular offset or per-angle jitter are configurable.

3.2 Classical baselines

Filtered back-projection (FBP) applies the standard ramp filter and inverse Radon transform. **Tikhonov regularization** solves

$$\hat{x} = \arg\min_x \tfrac{1}{2}\|Ax - y\|_2^2 + \tfrac{\lambda}{2}\|x\|_2^2 = (A^\top A + \lambda I)^{-1} A^\top y, \tag{3}$$

computed in closed form. **Total variation** solves

$$\hat{x} = \arg\min_x \tfrac{1}{2}\|Ax - y\|_2^2 + \lambda\,\mathrm{TV}(x) \tag{4}$$

via ADMM with an isotropic TV penalty and a finite-difference operator D, splitting $z = Dx$ and applying the standard shrinkage update on the split variable.

3.3 Learned prior: PnP proximal gradient

We train a DnCNN denoiser D_σ conditioned on the noise level σ (following the FFDNet-style noise-map input), on clean MNIST digits corrupted with Gaussian noise $\sigma \sim \mathcal{U}(0, \sigma_{\max})$, $\sigma_{\max} = 0.3$, using residual learning and an ℓ_2 loss. A single trained network is then used across all PnP experiments, with σ_d (the denoising strength argument passed at inference) treated as a tunable hyperparameter distinct from the training distribution's noise range.

Reconstruction proceeds by proximal-gradient iteration, replacing the proximal operator of an explicit regularizer with the learned denoiser:

$$x_{k+1} = D_{\sigma_d}\big(x_k - \eta\, A^\top (A x_k - y)\big), \qquad \eta = 1/L(A^\top A), \tag{5}$$

where $L(A^\top A)$ is the largest eigenvalue of $A^\top A$, estimated by power iteration. The iteration is initialized at the FBP reconstruction. Algorithm 1 summarizes the procedure.

Algorithm 1 PnP proximal gradient (PnP-PGD)

Require: measurements y, operator A, denoiser D, strength σ_d, step η, iterations K

$x_0 \leftarrow \mathrm{FBP}(y)$
for $k = 0, \dots, K-1$ **do**
  $g_k \leftarrow A^\top(Ax_k - y)$
  $x_{k+1} \leftarrow D_{\sigma_d}(x_k - \eta\, g_k)$
end for
return x_K

3.4 Evaluation protocol

Data are split into disjoint training, validation, and test subsets. All hyperparameters – λ for Tikhonov and TV, and (σ_d, K) for PnP – are selected by grid search maximizing mean validation PSNR, independently for each experimental condition (noise level or number of angles) in Experiments 1–2, and then evaluated once on the disjoint test set. For the robustness experiments (Section 4, E3–E4), hyperparameters are tuned once at a single nominal condition and then *frozen* across all mismatch/shift conditions, reflecting a realistic deployment scenario where a reconstruction pipeline is not retuned for every possible operating condition. We report PSNR and SSIM as mean with 95% confidence intervals over test images, and use a paired Wilcoxon signed-rank test (matched test images and noise realizations) to assess the significance of PnP vs. TV differences.

3.5 Data simulated on a finer grid

In E1–E5 the data are generated with the matrix A used for reconstruction (an "inverse crime" [8, 13]). For the experiments of Sections 4.5–4.9 we instead simulate measurements on a finer grid. The 28×28 image x is upsampled by an odd integer factor $f = 5$ (to 140×140) with bicubic interpolation and clipped to $[0, 1]$, projected with the same discrete Radon transform at the same angles, and every f adjacent detector bins are summed onto the 40-bin detector of the reconstruction operator, giving $y = \mathcal{S}(x) + n$ with $\mathcal{S} \neq A$. Reconstructions use A, and all metrics use the original 28×28 image as ground truth. The model error $\|\mathcal{S}(x) - Ax\|_2$ averages 0.09 at 20 angles (0.05 at 5 and 0.17 at 90 angles), i.e. 6% of the noise norm $\sigma\sqrt{m}$, $m = 40\, n_{\text{angles}}$, at $\sigma = 0.05$ and 31% at $\sigma = 0.01$. Because bicubic upsampling smooths the object, we also report a nearest-neighbour variant (object unchanged, model error 0.03) and a bilinear variant (model error ≈ 0.10, a stronger smoothing) as a sensitivity band. All hyperparameters are re-tuned on validation data simulated in the same way.

3.6 Weakly determined subspaces and prior-driven content

Let $A = U\Sigma V^\top$ be the SVD of the dense operator, with singular values $s_1 \geq \cdots \geq s_n$. For a threshold τ let $W_\tau = \mathrm{span}\{v_i : s_i < \tau s_1\}$ and let P_τ be the orthogonal projector onto W_τ. For 20 angles the operator has an exact null space of dimension 79 (singular values at machine precision), followed by a continuum of very small singular values with no gap; we therefore use $\tau = 0.05$ as the headline choice, giving $\dim W_{0.05} = 319$ of the 784 dimensions (and report $\tau = 0.10$ and the exact null space as sensitivity checks). A gradient step $x \mapsto x - \eta A^\top(Ax - y)$ never changes the component of x in the null space of A, so null-space content of a PnP reconstruction is supplied by

the denoiser alone. For a reconstruction $\hat{x}$ and ground truth x we report

$$\|\hat{x} - x\|_2^2 = \|P_\tau(\hat{x} - x)\|_2^2 + \|(I - P_\tau)(\hat{x} - x)\|_2^2 \quad \text{(orthogonal split)}, \tag{6}$$

$$\iota = \frac{\sum \|P_\tau \hat{x}\|_2^2}{\sum \|P_\tau x\|_2^2} \quad \text{(injection)}, \tag{7}$$

$$c = \frac{\sum \langle P_\tau \hat{x}, P_\tau x \rangle}{\sqrt{\sum \|P_\tau \hat{x}\|_2^2 \sum \|P_\tau x\|_2^2}} \quad \text{(alignment)}, \tag{8}$$

with sums over test images (pooled ratios). $\iota > 1$ means that the reconstruction places more energy in W_τ than the true images contain there; c measures how much of that content points in the right direction. These statistics cannot by themselves distinguish prior-driven structure from amplified noise, so we interpret them only in comparisons with identical noise, data and solver settings.

3.7 Denoiser seeds, noise range and matched-prior control

Besides the single MNIST denoiser of Section 3.3 we train, with the same architecture and optimiser (8 epochs, Adam, cosine schedule): (i) three MNIST denoisers with $\sigma_{\max} = 0.3$ (seeds 0–2); (ii) three MNIST denoisers with $\sigma_{\max} = 0.5$, which allows σ_d up to 0.5 and removes the grid-edge tuning at $\sigma = 0.2$ observed in E1; (iii) three denoisers trained on the Fashion-MNIST training split ($\sigma_{\max} = 0.3$). In the matched-prior control, all methods (Tikhonov, TV, PnP) are tuned on held-out Fashion-MNIST training images and tested on the Fashion-MNIST test split. The σ_d grid is extended to $\{0.005, \ldots, 0.3, 0.4, 0.5\}$ for all denoisers.

3.8 Lesion-insertion protocol

A lesion is an additive Gaussian blob, $x_{\text{les}} = \text{clip}_{[0,1]}\big(x + c\, e^{-\|p-p_0\|^2/(2\sigma_l^2)}\big)$, inserted at a random integer location p_0 of the central 16×16 region of a test image, with $\sigma_l \in \{0.5, 0.75, 1.25, 2.0\}$ px and contrast $c \in \{0.25, 0.5\}$ (eight lesion types). We reconstruct the lesion-present image and the lesion-absent original with independent noise realizations ($\sigma = 0.05$, 20 angles, crime-free data), using the same locations for every method. A lesion is *valid* if its inserted contrast in the ground truth is at least 50% of nominal (a lesion placed on a saturated stroke is clipped away); invalid cases are excluded for all methods, leaving 157–197 of 200 images per type. The observer is a location-known statistic, the *local contrast*: the mean of the image in a disc of radius $\rho = \lceil 1.5\sigma_l \rceil$ minus its mean in the surrounding annulus of width 3 px. We report

- the *paired AUC*: the fraction of images for which the statistic is larger in the lesion-present than in the lesion-absent reconstruction of the same base image (the background cancels; ties count one half);
- the *retention* $R = \mathrm{E}[t(\hat{x}_{\text{present}}) - t(\hat{x}_{\text{absent}})]/\mathrm{E}[t(x_{\text{les}}) - t(x)]$, which is 1 if the lesion contrast is preserved and below 1 if it is attenuated;
- the number of *false lesions* per lesion-absent image: local maxima of the contrast map of the reconstruction whose response is at least θ, half the response of the nominal lesion, and for which the ground truth has no contrast above $\theta/2$ within 2 px;
- the lesion's *observability*, the fraction of its energy outside $W_{0.05}$.

For the operating-point study (Section 4.9) we sweep 45 PnP settings ($\sigma_d \in \{0.02, 0.05, 0.1, 0.15, 0.2\}$, $K \in \{1, 2, 3, 5, 10, 25, 50, 100, 200\}$) and 9 values of the TV weight, choose operating points on 200

held-out *in-distribution* validation images (disjoint from the images used for PSNR tuning) and report them on the test sets, so that the choice is neither made on the evaluation images nor informed by the shifted distribution.

3.9 Matrix-free operator

Up to this point the forward operator A was stored as a dense 800×784 matrix, which allows exact adjoints, closed-form Tikhonov inverses and a full singular value decomposition, but does not scale to clinical image sizes. We therefore implemented a matrix-free operator that is never assembled. The forward projection embeds the image in a 40×40 canvas (the diagonal of the image, as in the dense operator), rotates it by each angle with bilinear interpolation about the pixel used by the reference implementation, and sums along the detector rows. The adjoint is the exact transpose of this discretisation, obtained by reverse-mode differentiation, so $\langle Ax, y\rangle = \langle x, A^\top y\rangle$ holds to floating-point precision by construction rather than approximately. The classical baselines use only forward and adjoint applications: Tikhonov by conjugate gradients on $A^\top A + \lambda I$, and TV by the same ADMM as before with the x-update solved by conjugate gradients. The singular subspaces used in Section 3.6 are obtained with a randomized SVD [7] (power iterations and oversampling, forward and adjoint products only); the error split in Section 4.6 then uses the well-determined subspace $\{i : s_i/s_1 \geq 0.05\}$ and its complement.

3.10 Poisson noise

Low-dose CT measures photon counts. With $p = Ax$ the (normalised) line integrals, the expected counts are $I = I_0 \exp(-\mu p)$ with I_0 incident photons per ray and attenuation scale $\mu = 2$; the counts are $N \sim \text{Poisson}(I)$ and the data used for reconstruction are $\hat{p} = -\mu^{-1} \log(\max(N, 1)/I_0)$. The noise in $\hat{p}$ is signal dependent, with variance approximately $e^{\mu p}/(I_0 \mu^2)$, and is larger along rays that pass through more material. We use $I_0 \in \{1000, 300, 100, 30, 10\}$, which gives an effective noise standard deviation of $0.018, 0.032, 0.056, 0.106, 0.198$ on the sinogram and therefore spans the Gaussian range $\sigma \in [0.02, 0.2]$ of the earlier experiments. The data are generated from the finer-grid object (Section 3.5, bicubic, factor 5), so the inverse crime remains removed. The reconstruction methods are unchanged and treat $\hat{p}$ with an unweighted least-squares data term (a Gaussian approximation); for PnP we also tested a variant with weights $w = N/I_0$ (an inverse-variance weighting up to a constant). All hyperparameters are tuned at each dose on 100 held-out MNIST validation images and frozen for every test set.

3.11 LoDoPaB-CT setting

LoDoPaB-CT [9] provides 362×362 slices with values $\mu/\mu_{\max} \in [0, 1]$ and simulated low-dose observations of 1000 parallel-beam angles on 513 detector bins. The official split has 35 820 training slices from 632 patients, 3 522 validation slices from 60 patients and 3 553 test slices from 60 patients; we verified that the three sets of patient identifiers are disjoint. We use 8 960 training slices (from the 632 training patients) to train the prior, 20 validation slices, one from each of 20 patients, for all tuning, and 40 test slices, one from each of 40 patients (the same 40 in every experiment), for all evaluation; the patient is therefore the unit of statistical analysis. The projector is a matrix-free parallel-beam operator with a unit-tested adjoint; its conventions (rotation sign, detector flip, image transposition, angle offset) were fixed by projecting the ground truth and comparing with the supplied observations over 16 variants: the selected one has a relative residual of 0.038 (correlation 0.9979), the best alternative 0.166, and the fitted pixel size is 1.0044 times the nominal 0.26 m/362. Sparse-view data use $K \in \{30, 60, 120\}$ evenly spaced angles out of the 1000.

3.12 Dose model and simulation

The observations follow $y = -\ln(N/N_0)/\mu_{\max}$ with $N \sim \text{Poisson}(N_0 e^{-\mu_{\max} Ax})$, $N_0 = 4096$ photons per ray, $\mu_{\max} = 81.35858\,\text{m}^{-1}$, and zero counts replaced by 0.1. This was checked on the data: the counts recovered from the observations are integers in 100% of the bins for $N_0 = 4096$ (and 8192), against 9.7–10.4% for perturbed values of $\mu_{\max}$ (chance level 10%) and 24.7% and 49.9% for $N_0 = 1024$ and 2048. The dispersion index $\overline{(N-\lambda)^2/\lambda}$ around the counts expected from the ground truth is 1.06, 1.25 and 1.81 for expected counts in $[20, 200)$, $[200, 1000)$ and $[1000, 5000)$; values above one include discretisation and ground-truth error, so the real data may contain somewhat more variance than the simulation. Lower doses $f \in \{1, 0.25, 0.1\}$ (4096, 1024 and about 410 photons per ray) are obtained from the real observations by exact binomial thinning of the recovered counts (a thinned Poisson variable is Poisson with f times the mean), which is used for *tuning*. For *evaluation* the data are simulated without the inverse crime: the slice is up-sampled by a factor two (bicubic), projected on the 724×724 grid with half-size pixels and 1026 detector bins, pairs of bins are averaged, and Poisson noise is added exactly as above. Table 13 shows that reconstructions from simulated and from real data of the same slices have the same TV PSNR within 0.15 dB in all nine conditions.

3.13 Methods and tuning

FBP-Hann: filtered back-projection with a Hann window whose cut-off (a fraction of Nyquist) is tuned. *TV*: isotropic total variation with the box constraint $[0, 1]$, solved by the primal–dual algorithm of Chambolle and Pock [4] for 150 iterations from the FBP-Hann image; *TV-long* is the same with 300 iterations and its own tuned weight, a control for the extra iterations that PnP performs. *PnP-const*: $x_{k+1} = D_{\sigma_d}\big(x_k - \eta A^\top (Ax_k - y)\big)$, $\eta = 1/L$, started from the FBP-Hann image, with (σ_d, k) tuned; *PnP-TVinit*: the same started from the TV solution. D_σ is a noise-level-conditioned DnCNN (8 layers, 64 channels, σ as a second input channel, residual output) trained for 15 000 steps (batch 16, random 128×128 crops and flips, $\sigma \sim U(0, 0.1)$, Adam, cosine schedule); on the 20 validation slices it improves the PSNR of images with additive Gaussian noise from 33.98, 26.02, 20.00 dB to 41.98, 38.45, 35.53 dB at $\sigma = 0.02, 0.05, 0.1$. Every method and condition is tuned separately on the 20 validation slices (real observations, thinned and angle-subsampled) by maximising the mean PSNR (peak 1, reconstructions clipped to $[0, 1]$); whenever the optimum falls on the boundary of a grid, the grid is extended (up to twice; cut-off $\times 0.75$ or $+0.1$, TV weight $\times 3$ or $\div 3$, σ_d $\times 1.6$ or $\div 1.6$). Table 14 lists the final choices; two optima remain on a boundary: $\sigma_d = 0.0039$ of PnP-TVinit at $K = 120$, full dose (the lower end after two extensions, i.e. almost no denoising) and $k = 1$ of PnP-TVinit at $K = 120$, dose 0.1 (the smallest iteration count allowed). In these two cells PnP-TVinit is essentially TV followed by a few data-consistency steps.

3.14 Lesion observer

The protocol of Section 3.8 is applied to 362×362 slices. Lesions are additive Gaussian blobs, clipped to $[0, 1]$, with $\sigma_l \in \{2, 4\}$ px (1.4 and 2.9 mm at 0.72 mm per pixel) and contrast $c = 0.06$ on the $\mu/\mu_{\max}$ scale (in the step at $K = 60$ also $c = 0.03$). If water is taken to be about 0.25 on this scale, which is an assumption, $c = 0.06$ and 0.03 correspond to roughly 240 and 120 HU; the lesions are small and noisy at low dose but are not low-contrast in the clinical sense. Four lesions of one type are inserted into each slice at random positions at least $10\sigma_{l,\max} + 4 = 44$ px apart, inside the body eroded by 22 px (CT slices) or in the central square (phantoms); the lesion-free and lesion-present images are reconstructed from independent noise realisations. A lesion is valid if its inserted response is at least half the nominal one. The observer is the location-known, zero-sum

local-contrast statistic of Section 3.8 (disc of radius $\lceil 1.5\sigma_l \rceil$ minus a 3-px annulus), and the detection score is the paired AUC pooled over valid lesions (up to $40 \times 4 = 160$ lesions per lesion type on CT slices, $20 \times 4 = 80$ on phantoms). A *false blob* is a local maximum (in a 5×5 window) of the response map of a lesion-*free* reconstruction whose response is at least $m\, r_{\mathrm{nom}}$ (r_{nom}: response to the nominal lesion on an empty background; $m \in \{0.5, 0.75, 1, 1.5\}$) and for which the truth has no response above half of $0.5\, r_{\mathrm{nom}}$ within 2 px. $m = 1$ and 1.5 count structures that a reader of this statistic would take for at least a full lesion. The statistic measures lesion-like response at the scale of the kernel; we did not run a reader study, so "convincing" is shorthand for this response.

3.15 Statistics

All comparisons are paired: two methods are evaluated on the same patients and the same simulated data, and the difference is summarised by its mean over patients with a 95% percentile bootstrap interval over patients (2 000 resamples in the $K = 60$ lesion experiments, 5 000 elsewhere), with the same resamples for both methods. Lesion types are pooled with equal weight; per-type rows are exploratory. For the shift comparison, the CT and phantom differences are bootstrapped independently (different images) and their difference is the *shift effect*. No correction for multiplicity is applied, so we report the numbers of cells with significant positive and negative differences and read the sign pattern across conditions rather than any single interval: with 80 per-type intervals at the 5% level about four would exclude zero by chance, whereas 25 and 35 did. Each condition uses one noise realisation, which limits the precision of a single cell (for AUC about ± 0.05); the 20 phantoms are the same in all conditions, so conditions are not independent.

3.16 Analysis history

The analysis was developed in steps, and some of them changed the conclusions. We list them because they affect how the final numbers should be read. (i) A first comparison at $K = 60$ (tuning on 6 validation slices, Hann cut-off 0.2) found PnP 0.85 and 0.33 dB below TV at full and quarter dose. Re-tuning on 20 slices (cut-off 0.3 at full dose) raised PnP-const by 1.06 dB (30.68 → 31.74) and 0.32 dB (29.40 → 29.72) but TV by only 0.38 and 0.09 dB, and the gap became −0.17 and −0.10 dB (Table 15); the initial gap was an artefact of the starting image and the small tuning set, and we do not use it. (ii) The lesion experiments were first tuned on 10 slices with narrow grids; they were repeated with 20 slices, extended grids and a separately tuned TV-long. At full dose the parameters of FBP, TV, TV-long and PnP-const did not change (the PSNR of PnP-TVinit changed by 0.07 dB) and no pooled verdict changed; at quarter dose one of 20 pooled verdicts changed (PnP-TVinit − TV-long AUC, from significant to not significant). (iii) Paired statistics, the threshold sweep and the false-blob definition at several strengths were added after seeing the first lesion results, and the dose × angle grid and the shift extension to $K = 30, 120$ were run afterwards; none of this was pre-specified. (iv) Edge-of-grid optima and the coarse TV grid (factor three) are described in Section 3.13 and in the Limitations.

4 Experiments

4.1 E1–E2: In-distribution accuracy under noise and undersampling

Table 1 reports test-set PSNR for representative conditions (20 projection angles, varying measurement noise σ; and fixed $\sigma = 0.05$, varying number of angles).

Table 1: Test PSNR (dB) under varying noise and angular undersampling. Best method per row in bold. PnP–TV column reports the paired Wilcoxon test outcome. Data generated with the reconstruction operator itself ("inverse crime"); see Table 4 and Table 5 for data simulated on a finer grid.

Condition	FBP	Tikhonov	TV	PnP-DnCNN	PnP − TV
σ=0, 20 angles	24.2	28.9	**36.0**	29.6	-6.4 ($p < 10^{-30}$)
σ=0.01, 20 angles	23.1	24.1	29.0	29.1	$+0.07$ (n.s., $p = 0.27$)
σ=0.05, 20 angles	15.5	17.1	20.2	**24.0**	$+3.85$ ($p < 10^{-30}$)
σ=0.1, 20 angles	10.4	14.7	16.9	**19.9**	$+3.0$ ($p < 10^{-30}$)
σ=0.2, 20 angles	6.6	12.9	14.3	**15.3**	$+1.0$ ($p < 10^{-10}$)
5 angles, σ=0.05	9.1	14.2	15.5	**17.7**	$+2.15$ ($p < 10^{-25}$)
10 angles, σ=0.05	12.1	15.9	18.4	**21.4**	$+3.05$ ($p < 10^{-30}$)
90 angles, σ=0.05	20.6	20.0	24.1	**27.1**	$+3.04$ ($p < 10^{-30}$)

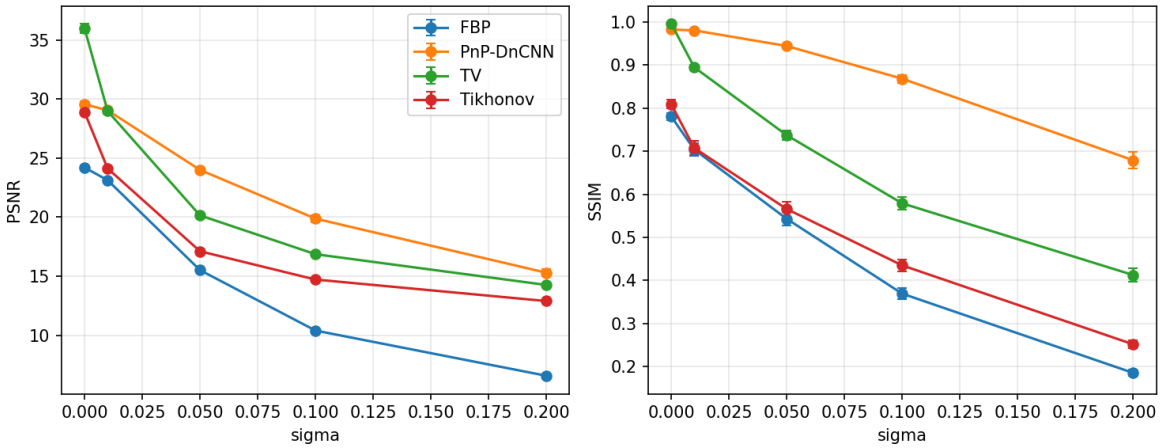


Figure 1: Reconstruction accuracy (PSNR, left; SSIM, right) vs. measurement noise level σ, for all four methods at a fixed 20 projection angles. TV is the strongest method at $\sigma \leq 0.01$; PnP-DnCNN overtakes all classical baselines for $\sigma \geq 0.05$. Error bars denote 95% confidence intervals over 200 test images.

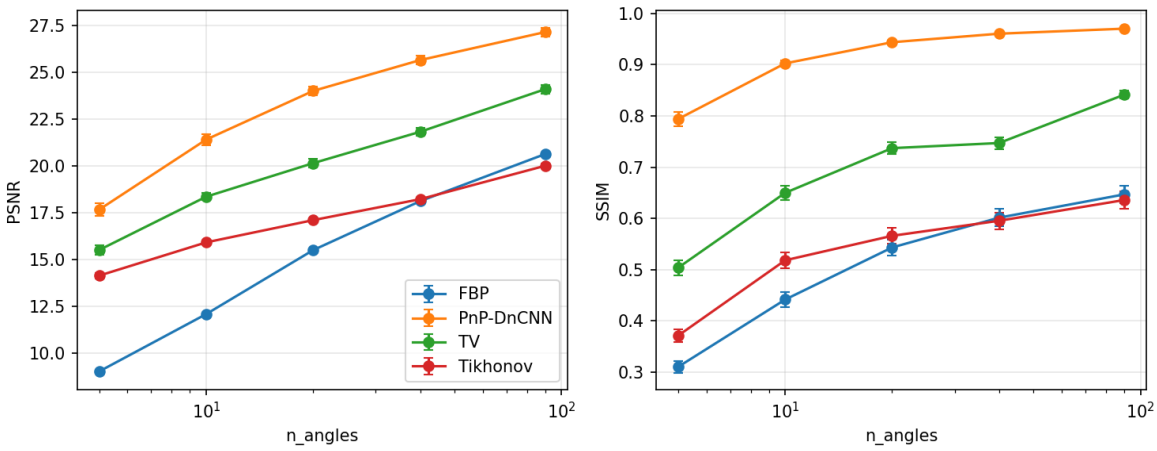


Figure 2: Reconstruction accuracy (PSNR, left; SSIM, right) vs. number of projection angles, for all four methods at a fixed measurement noise $\sigma = 0.05$. PnP-DnCNN is the top performer across the full range tested (5 to 90 angles); its advantage over TV is largest at 20–40 angles ($\approx +3.9$ dB) and smaller at the extremes (+2.2 dB at 5 angles, +3.0 dB at 90 angles). Error bars denote 95% confidence intervals over 200 test images.

TV is the strongest classical baseline throughout, consistent with MNIST digits being close to piecewise-constant. The learned prior overtakes TV as soon as the problem departs from the near noise-free, moderately-sampled regime. The PnP–TV gap peaks at about +3.9 dB, at $\sigma = 0.05$ with 20–40 angles; it is smaller at the extremes (+2.2 dB at 5 angles, +3.0 dB at 90 angles, +1.0 dB at $\sigma = 0.2$), with a corresponding SSIM improvement (0.74 for TV to 0.94 for PnP at $\sigma = 0.05$, 20 angles; full SSIM table in the project repository). At $\sigma = 0$ the learned prior underperforms TV by 6.4 dB and the two are statistically indistinguishable at $\sigma = 0.01$; Section 4.5 shows that the size of these two low-noise gaps depends on how well the data model matches the reconstruction model.

4.2 E3: Forward-model mismatch

Measurements are generated with the true 20-angle operator; reconstruction uses a perturbed operator with either a constant angle offset (rigid misregistration) or independent per-angle jitter, up to 4°, with all hyperparameters frozen at their $\sigma = 0.05$, 20-angle values. Table 2 summarizes the extremes of the tested range.

Table 2: Test PSNR (dB) under forward-model mismatch, hyperparameters frozen at nominal values.

Mismatch	TV (0° → 4°)	PnP (0° → 4°)	TV loss	PnP loss
Constant offset	20.2 → 18.6	24.0 → 20.0	1.6 dB	4.0 dB
Per-angle jitter	20.2 → 19.2	24.0 → 22.2	1.0 dB	1.8 dB

PnP remains more accurate than TV in absolute terms throughout the tested range, but degrades substantially faster in relative terms, particularly under a constant angular offset. Its advantage over TV narrows from +3.85 dB at zero mismatch to +1.48 dB at 4° offset.

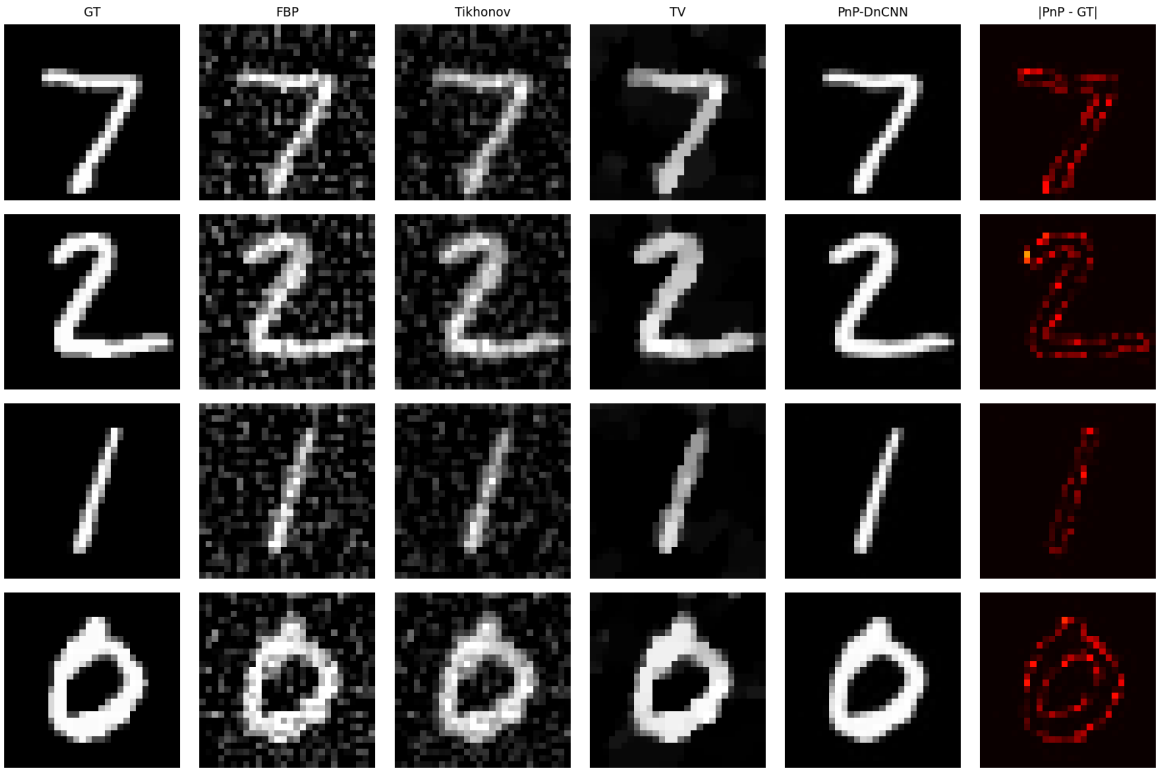


Figure 3: Qualitative in-distribution reconstructions at $\sigma = 0.05$, 20 angles. Each row is one test digit; columns show the ground truth, the three classical baselines, PnP-DnCNN, and the absolute error map $|\hat{x}_{\text{PnP}} - x|$ (brighter = larger error). FBP and Tikhonov retain visible measurement noise; TV produces flat, blocky regions; PnP-DnCNN recovers the sharpest, most faithful strokes. The error maps show that residual error is concentrated almost entirely along stroke boundaries, with near-zero error in flat background and interior regions.

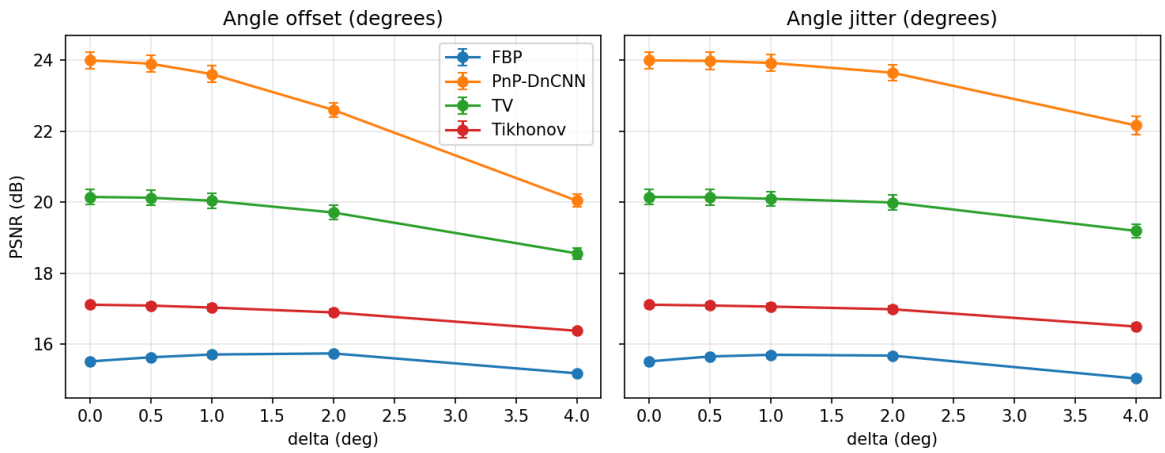


Figure 4: Test PSNR under forward-model mismatch, as a function of angular error δ (degrees), for a constant angle offset (left) and independent per-angle jitter (right). All hyperparameters are frozen at their nominal ($\delta = 0$) values and not retuned per mismatch level. PnP-DnCNN remains the most accurate method throughout the tested range, but degrades faster than TV, particularly under the systematic offset: its advantage over TV narrows from +3.85 dB at $\delta = 0$ to +1.48 dB at $\delta = 4°$. Error bars denote 95% confidence intervals over 200 test images.

4.3 E4: Distribution shift

The denoiser is trained exclusively on MNIST digits. We evaluate the full frozen pipeline (nominal $\sigma = 0.05$, 20 angles) on Fashion-MNIST, EMNIST letters, and a resized Shepp-Logan phantom (single image, 20 independent noise draws). Table 3 and Figure 6 summarize the results.

Table 3: Test PSNR (dB) under train–test distribution shift, frozen pipeline.

Test set	TV	PnP-DnCNN	PnP − TV
MNIST (in-distribution)	20.3	**24.0**	+3.7
EMNIST letters (near shift)	20.1	**22.0**	+1.9
Fashion-MNIST (far shift)	**19.6**	16.5	−3.1
Shepp-Logan phantom (far shift)	**19.2**	15.0	−4.2

The learned prior generalizes to EMNIST letters, which share digits' stroke-based statistics, but underperforms TV on both far-shift test sets. Qualitatively (Figure 6), the failure mode is not generic blur but hallucination: the denoiser imposes plausible-looking but incorrect digit/stroke-like structure on inputs unlike its training distribution.

4.4 E5: Why PnP selects few iterations

Validation tuning consistently selects a small number of PnP iterations (5–25 for ≥ 10 angles and $\sigma \leq 0.2$, and 100–400 for the 5-angle condition). Figure 7 shows PSNR and the data-fidelity residual $\|Ax_k - y\|$ as a function of iteration count, for several values of σ_d, on 50 test images. PSNR rises rapidly and peaks within the first 5–15 iterations, then *decays* to a lower plateau (e.g. $\sigma_d = 0.1$: peak ≈ 24.0 dB near iteration 10, plateau ≈ 22.0 dB by iteration 400). Over the same range, the residual does not decrease; it rises and plateaus, more steeply for larger σ_d. This indicates that the

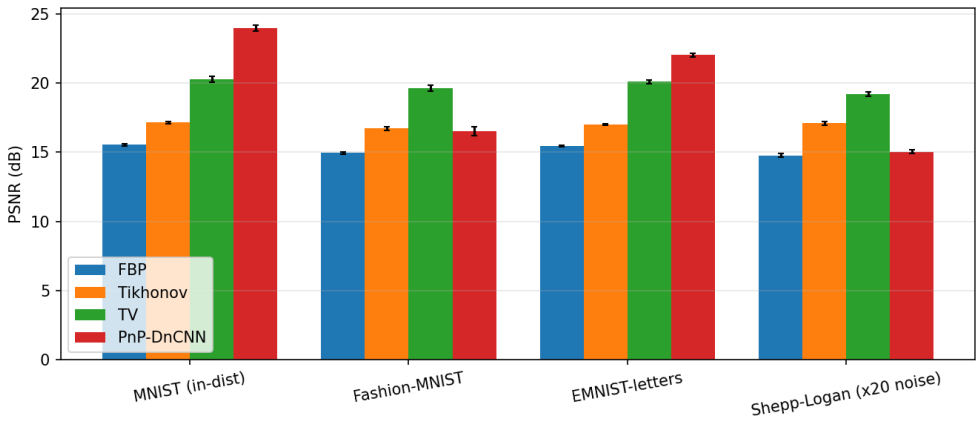


Figure 5: Test PSNR under train–test distribution shift, for all four methods, with the reconstruction pipeline frozen at its nominal ($\sigma = 0.05$, 20 angles) hyperparameters. PnP-DnCNN leads on in-distribution MNIST and on EMNIST letters (a near-shift domain sharing digits' stroke-like statistics), but falls below TV and Tikhonov on Fashion-MNIST and the Shepp–Logan phantom (by 3.0 and 0.2 dB on Fashion-MNIST, by 4.2 and 2.1 dB on the phantom); it stays above FBP on Fashion-MNIST (16.6 vs. 14.9 dB) and only marginally so on the phantom (15.0 vs. 14.8 dB). TV is the most consistent performer across all four test sets. Error bars denote 95% confidence intervals; for the Shepp-Logan phantom, the interval reflects variation across 20 independent noise draws of a single image rather than variation across distinct images.

iteration does not converge to a fixed point that is simultaneously data-consistent and maximally accurate: the denoiser's implicit prior increasingly dominates the data-fidelity term as iterations progress, without a corresponding gain in accuracy. Early stopping is therefore not an artifact of a coarse hyperparameter grid but an active, necessary part of the regularization.

4.5 E6: Sensitivity to the inverse crime

Table 4 compares the PnP–TV gap under four data models. For $\sigma \geq 0.05$ the model error is at most 7% of the noise norm and the in-distribution PnP–TV gaps change by at most 0.3 dB (bicubic data), so the conclusions of E1–E2 are unaffected. At $\sigma \leq 0.01$ they are not: TV remains ahead at $\sigma = 0$ in every variant, but by 5.9 dB with the nearest-neighbour object and by only 1.0 dB with the smoother bilinear object, and at $\sigma = 0.01$ PnP's lead ranges from +0.07 to +1.76 dB. The low-noise ranking is therefore a function of the forward-model mismatch rather than a single number, which also explains the sign of the $\sigma = 0$ result in E1. The out-of-distribution deficits of E4 persist in every variant (Fashion-MNIST: -2.8 to -3.9 dB; Shepp–Logan: -4.2 to -5.4 dB). We use bicubic data, the intermediate choice, as the default crime-free setting; Table 5 repeats the E1–E2 comparison with it.

4.6 E7: Error decomposition along the singular vectors

Table 6 reports the decomposition of Section 3.6 at $\sigma = 0.05$ and 20 angles ($\dim W_{0.05} = 319$). In distribution, PnP has about 0.4–0.5 times TV's error energy in *both* the weakly determined subspace and its complement (null-space ratio 0.48, range-space ratio 0.40) and a better-aligned completion of the unobserved directions ($c = 0.76$ versus 0.45): the prior's advantage is not confined to filling in unobserved directions. Under shift both ratios exceed 1 on Fashion-MNIST (1.27 and 2.32) and on

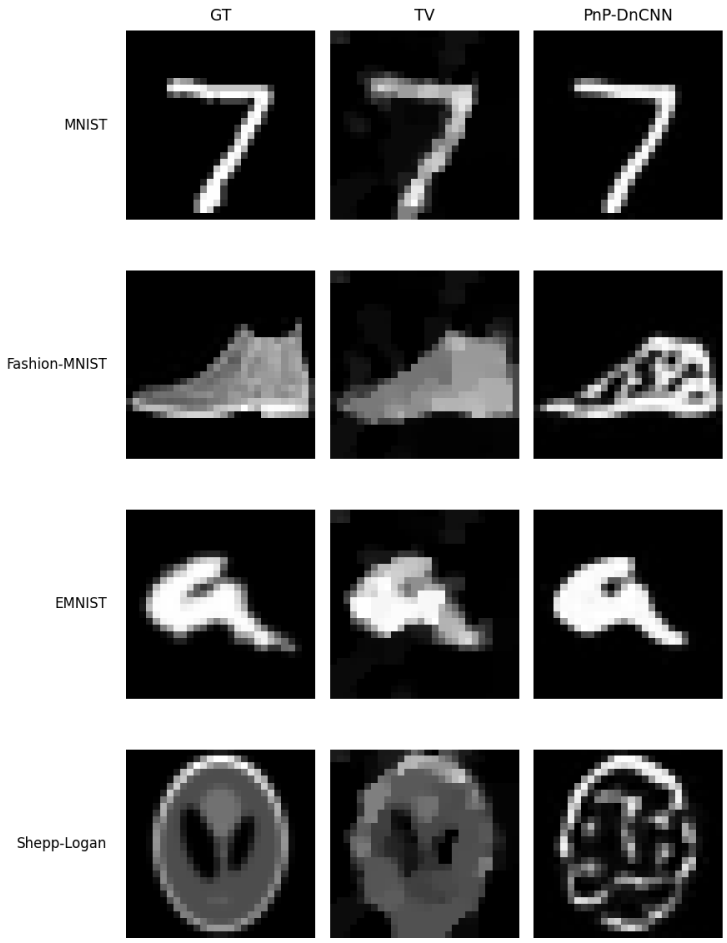


Figure 6: Qualitative reconstructions under distribution shift. The learned prior recovers structure well on MNIST and EMNIST (both stroke-like), but imposes hallucinated digit/stroke-like structure on Fashion-MNIST and the CT phantom, both structurally unlike the training distribution.

Table 4: PnP−TV PSNR gap (dB), 20 angles, 200 test images, for four data models. "Model error" is the mean $\|\mathcal{S}(x) - Ax\|_2$; for comparison the noise norm $\sigma\sqrt{m}$ is 0.28, 1.41, 2.83, 5.66 for $\sigma = 0.01$, 0.05, 0.1, 0.2. Hyperparameters are re-tuned for each data model.

Data model	Model error	$\sigma = 0$	0.01	0.05	0.1	0.2
Same operator (inverse crime)	0	−6.43	+0.07	+3.86	+3.01	+1.03
Fine grid, nearest	0.03	−5.86	+0.16	+3.86	+3.01	+1.02
Fine grid, bicubic (default)	0.09	−3.29	+1.02	+4.02	+3.10	+1.08
Fine grid, bilinear	0.10	−0.97	+1.76	+3.94	+2.87	+0.85

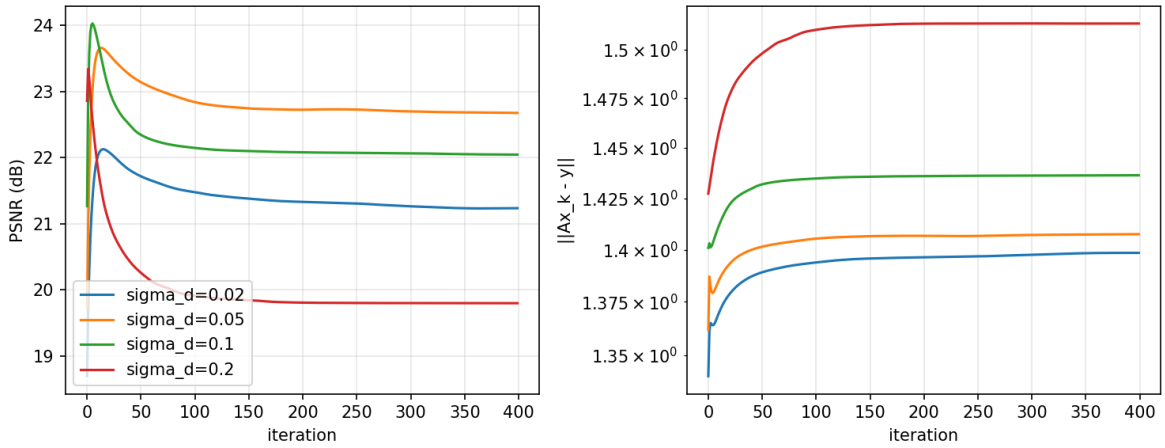


Figure 7: PnP-PGD convergence behavior for several denoising strengths σ_d (50 test images, $\sigma = 0.05$, 20 angles). Left: PSNR vs. iteration, peaking early then decaying. Right: data-fidelity residual vs. iteration, rising and plateauing rather than decreasing.

Table 5: As Table 1, with measurements simulated on the finer grid (bicubic, $f = 5$); mean test PSNR (dB) over 200 images. Significance tests are provided with the repository results.

Condition	FBP	Tikhonov	TV	PnP-DnCNN	PnP−TV
$\sigma = 0$, 20 angles	24.3	27.4	33.2	29.9	−3.29
$\sigma = 0.01$, 20 angles	23.2	23.9	28.4	29.4	+1.02
$\sigma = 0.05$, 20 angles	15.5	17.1	20.1	24.1	+4.02
$\sigma = 0.1$, 20 angles	10.4	14.7	16.8	19.9	+3.10
$\sigma = 0.2$, 20 angles	6.6	12.9	14.3	15.3	+1.08
5 angles, $\sigma = 0.05$	9.1	14.1	15.5	17.8	+2.36
10 angles, $\sigma = 0.05$	12.1	15.9	18.2	21.5	+3.29
40 angles, $\sigma = 0.05$	18.1	18.2	21.8	25.8	+3.97
90 angles, $\sigma = 0.05$	20.7	20.0	24.1	27.3	+3.24

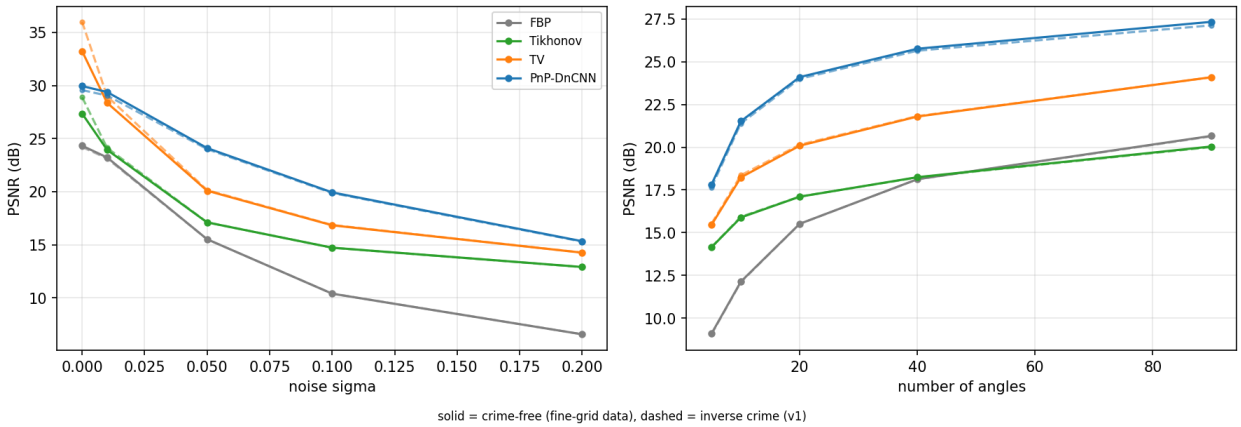


Figure 8: PSNR versus noise level (left) and number of angles (right) for data simulated on the bicubic finer grid (solid) and with the inverse crime (dashed). The curves coincide for $\sigma \geq 0.05$ and separate only at $\sigma \leq 0.01$, where TV loses most.

the phantom (2.59 and 2.64), so the failure is not confined to the weakly determined subspace either, and the largest part of the excess error lies in its complement. The injection statistic shows the hallucination signature: on the phantom PnP places 3.3 times the true energy in $W_{0.05}$ (TV: 0.44) with alignment 0.23, while on MNIST its 0.76 comes with alignment 0.76. Within the iteration, the contraction of direction i by the gradient step after K iterations is $(1-(s_i/s_1)^2)^K$, so at $K=5$ a direction with $s_i = 0.1\,s_1$ retains 95% of its initial error, $s_i = 0.3\,s_1$ 62% and $s_i = 0.5\,s_1$ 24%: the tuned iteration leaves a large share of directions to the denoiser. On the shifted sets no setting of (σ_d, K) recovers TV's accuracy: the best oracle setting still has 1.5–1.9 times TV's error and the error grows with K (to 3.7–4.6 times at $K = 400$; bilinear fine-grid data).

Table 6: Error decomposition at $\sigma = 0.05$, 20 angles, bicubic fine-grid data, $W_{0.05}$ (dim = 319). Ratios are PnP/TV (below 1: PnP has less error); ι is the injection and c the alignment statistic of Section 3.6.

	error ratio PnP/TV			ι		c	
Test set	in W	in $W^\perp$	total	TV	PnP	TV	PnP
MNIST (in-dist.)	0.48	0.40	0.42	0.58	0.76	0.45	0.76
EMNIST letters	0.70	0.64	0.65	1.22	1.55	0.43	0.67
Fashion-MNIST	1.27	2.32	2.05	0.29	0.80	0.36	0.37
Shepp–Logan	2.59	2.64	2.63	0.44	3.27	0.09	0.23

4.7 E8: Denoiser seeds and matched-prior control

Table 7 reports the shift experiment with three denoisers per configuration (bicubic data, $\sigma = 0.05$, 20 angles; all methods re-tuned on the validation distribution of the respective prior). Seed-to-seed variation is at most 0.3 dB, an order of magnitude below the effects of interest: on Fashion-MNIST the MNIST-trained prior is 3.10 ± 0.30 dB below TV (range -3.43 to -2.83) and on the phantom 4.33 ± 0.12 dB below. Training with $\sigma_{\max} = 0.5$ instead of 0.3 does not change these results (Fashion-MNIST -3.12, phantom -3.94 dB) and, at $\sigma = 0.2$, the tuned σ_d is 0.3 for all six MNIST denoisers although values up to 0.5 were available, so the grid-edge tuning in E1 was not an artefact; the wider-range denoiser is slightly better at high noise ($+1.44 \pm 0.11$ versus $+0.97 \pm 0.11$ dB over TV at $\sigma = 0.2$, ranges not overlapping) and slightly worse at $\sigma = 0.05$.

The matched-prior control isolates the cause of the reversal: a prior trained on Fashion-MNIST beats TV on Fashion-MNIST by $+2.21 \pm 0.01$ dB (error ratio 0.63), although less than the MNIST prior on MNIST (+3.84 dB), since Fashion-MNIST images are less sparse. The control is clean: TV's PSNR is identical for all priors on every test set, i.e. tuning on Fashion-MNIST validation data selects the same TV weight. Transfer is asymmetric: the Fashion-trained prior still beats TV on MNIST (+1.38 dB) and ties it on the phantom (-0.18 dB, error ratio 1.05), whereas the MNIST-trained prior loses by 4.3 dB on the phantom. The injection statistic on the phantom follows the prior: 2.8–3.4 for MNIST-trained denoisers versus 0.30–0.35 for Fashion-trained ones, with identical noise and solver settings. Whether the broader prior helps because it is richer or because Fashion-MNIST resembles the phantom more cannot be decided with two training distributions.

4.8 E9: Lesion insertion

Table 8 and Figure 11 summarise the lesion experiment (three denoisers per prior, protocol of Section 3.8). Three observations hold across lesion types.

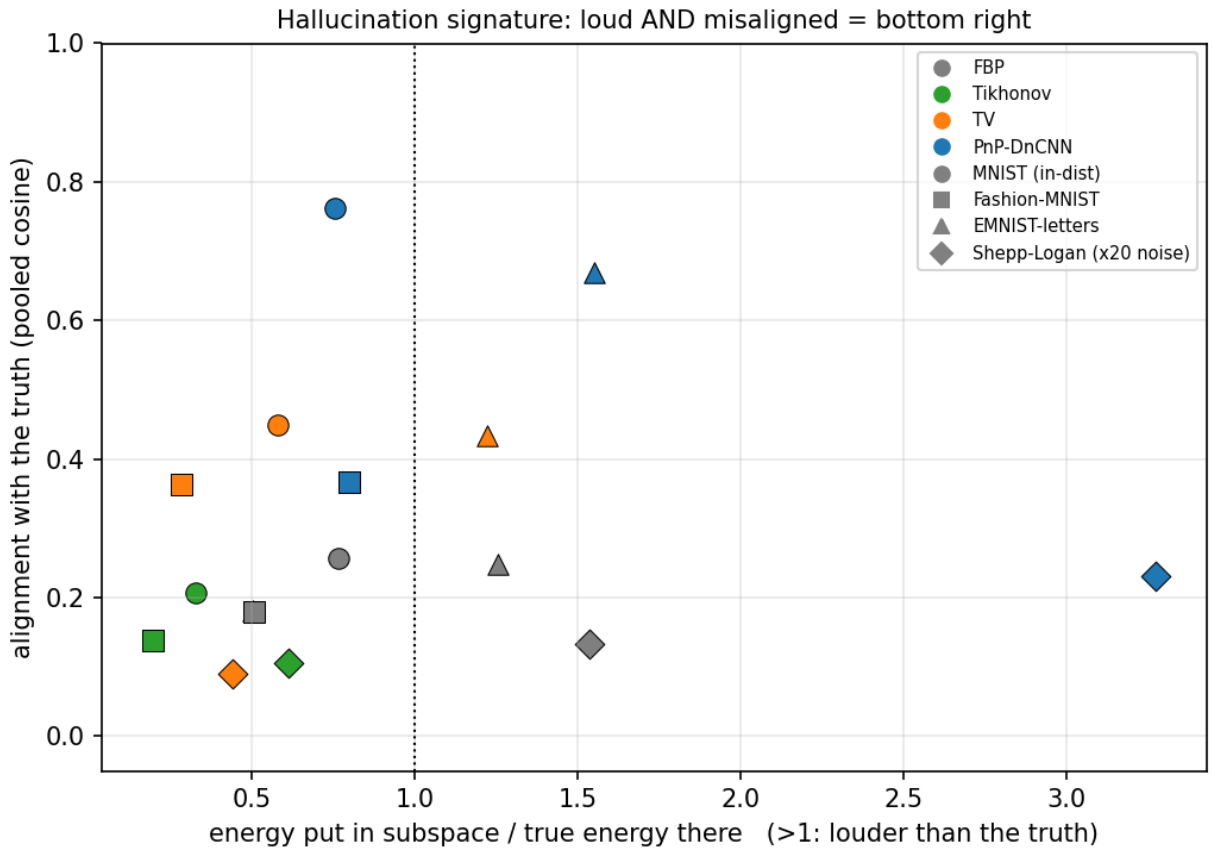


Figure 9: Hallucination signature (injection versus alignment in the weakly determined subspace) for TV and PnP-DnCNN on the four test sets (bicubic fine-grid data).

Table 7: Distribution shift with three denoisers per prior (mean test PSNR in dB over seeds; gap = PnP−TV, mean ± s.d. over seeds). Bicubic fine-grid data, $\sigma = 0.05$, 20 angles; each prior's pipeline is tuned on the validation set of its own training distribution.

		MNIST-trained		Fashion-trained	
Test set	TV	PnP	gap	PnP	gap
MNIST	20.21	24.05	$+3.84 \pm 0.03$	21.60	$+1.38 \pm 0.09$
EMNIST letters	20.28	22.07	$+1.79 \pm 0.09$	20.73	$+0.44 \pm 0.09$
Fashion-MNIST	19.42	16.32	-3.10 ± 0.30	21.63	$+2.21 \pm 0.01$
Shepp–Logan	19.28	14.96	-4.33 ± 0.12	19.10	-0.18 ± 0.08

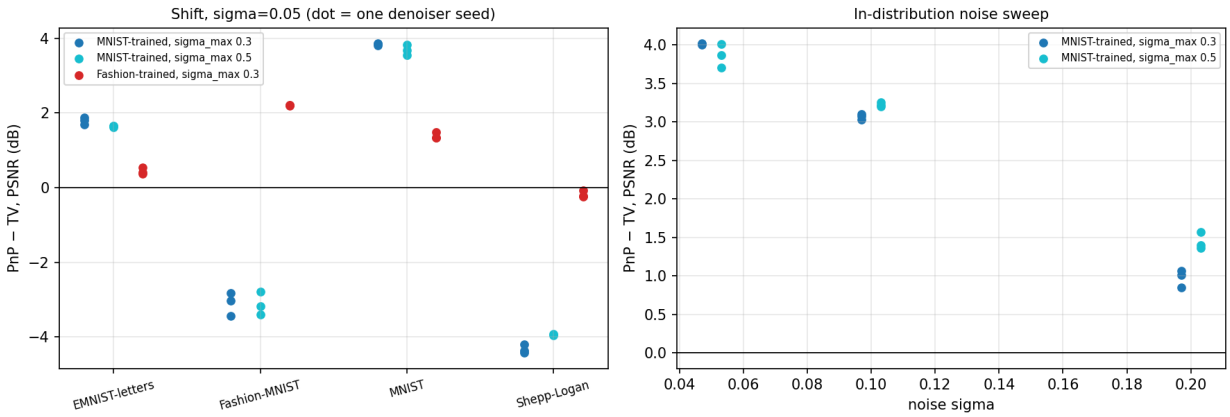


Figure 10: Left: PnP−TV gap for each denoiser seed (dots) on the four test sets, for three priors. Right: gap in the in-distribution noise sweep for MNIST-trained denoisers with $\sigma_{\max} = 0.3$ and 0.5.

TV detects small lesions better than PnP, in distribution and for a matched prior. For the MNIST-trained prior on MNIST the paired AUC of PnP is 0.04–0.23 below TV's (for example 0.69 versus 0.91 for $\sigma_l = 0.75$, $c = 0.5$), although PnP is about 4 dB better in PSNR on the same images; for the matched Fashion-trained prior the deficit is 0.03–0.17. The two methods tie only for the largest high-contrast lesion. FBP and Tikhonov also detect small lesions better than PnP (paired AUC 0.92 at $\sigma_l = 0.75$, $c = 0.5$ on MNIST), at the cost of many false blobs (up to 6 per image for FBP).

In distribution PnP attenuates lesions and invents none. Its retention is 0.12–0.78 for the MNIST prior on MNIST and 0.16–0.49 for the matched prior (TV: 0.29–0.88 and 0.38–0.83), and it produces at most 0.01 false lesions per image against up to 0.26 for TV. The reconstructions are cleaner than TV's but the inserted lesion is often absent (Figure 12), so a visually convincing image does not imply a faithful one.

Under prior mismatch PnP invents structure. For the MNIST-trained prior on Fashion-MNIST the retention is 0.78–1.04 (the lesion survives) but there are 0.20–0.66 false lesions per image against ≤ 0.20 for TV, and the paired AUC remains below TV's by 0.05–0.16. The matched Fashion-trained prior on the same images produces at most 0.01. Lesion observability (Section 3.8) does not explain these differences: the observability terciles formed within each lesion type span only 0.87–0.95 on average, and PnP's AUC does not increase with observability; smaller lesions are harder for all methods, but this is confounded with size.

4.9 E10: Can re-tuning recover lesion detectability?

The PSNR-tuned operating point of PnP is not necessarily the best one for lesion detection, so we sweep operating points (Section 3.8) and compare three rules for PnP—PSNR-tuned, best paired AUC on the selection set ("lesion"), and best paired AUC among points whose selection-set PSNR is at least that of tuned TV ("constrained")—with TV. Table 9 and Figure 13 show one denoiser per prior. For the MNIST-trained prior in distribution, the constrained point keeps a +0.7 dB PSNR advantage over TV (the PSNR-tuned advantage was +4.0 dB) but its paired AUC is still 0.06 below TV's, and its false-lesion rate rises to 12 times TV's; the lesion-tuned point matches TV's detection but has lower PSNR than TV and 60 times TV's false-lesion rate (0.32 versus 0.005 per image). For the matched Fashion-trained prior the constrained point keeps +1.0 dB over TV,

Table 8: Lesion insertion, mean over three denoisers per prior (200 test images, $\sigma = 0.05$, 20 angles, bicubic data), for two of the eight lesion types (σ_l/c = lesion size in px / contrast). pAUC: paired AUC; Ret.: retention; FL: false lesions per lesion-absent image. All methods use their PSNR-tuned settings.

		pAUC		Ret.		FL	
Prior → test set	σ_l/c	TV	PnP	TV	PnP	TV	PnP
MNIST → MNIST (in-dist.)	0.75/0.5	0.91	0.69	0.59	0.28	0.005	0.000
	1.25/0.25	0.90	0.69	0.62	0.30	0.005	0.000
MNIST → Fashion (shifted)	0.75/0.5	0.91	0.82	0.63	0.99	0.030	0.430
	1.25/0.25	0.93	0.76	0.59	0.87	0.030	0.428
Fashion → Fashion (in-dist.)	0.75/0.5	0.91	0.75	0.63	0.22	0.030	0.005
	1.25/0.25	0.93	0.80	0.59	0.23	0.030	0.005
Fashion → MNIST (shifted)	0.75/0.5	0.91	0.75	0.59	0.35	0.005	0.000
	1.25/0.25	0.90	0.78	0.62	0.36	0.005	0.000

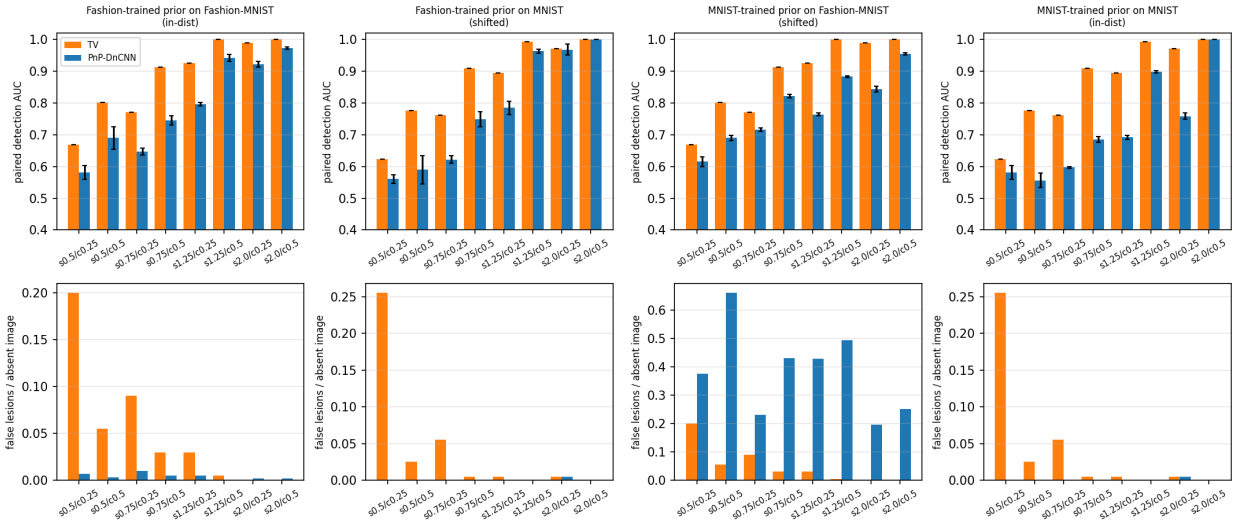


Figure 11: Paired detection AUC (top) and false lesions per lesion-absent image (bottom) for TV and PnP-DnCNN in the four prior/test-set combinations and the eight lesion types (mean over three denoisers; bars: s.d. over seeds).

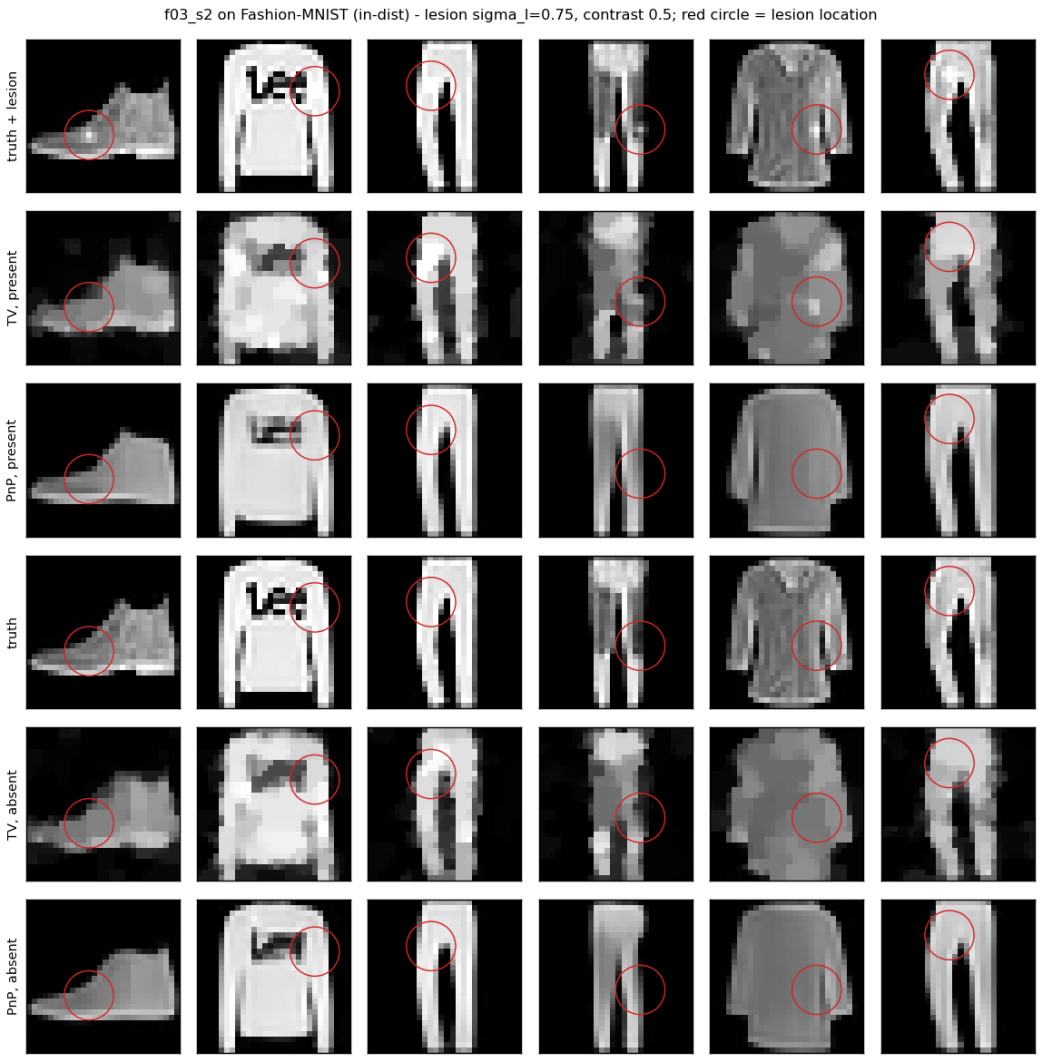


Figure 12: Matched Fashion-trained prior on Fashion-MNIST ($\sigma_l = 0.75$, $c = 0.5$). Rows: ground truth with lesion, TV and PnP reconstructions of the lesion-present image; ground truth, TV and PnP reconstructions of the lesion-absent image. The red circle marks the lesion location. PnP produces the cleaner images but removes the lesion.

with a paired AUC within 0.02–0.03 of TV's (about one standard error) and 0.085 false lesions per image against 0.03 for TV. Under shift, operating points chosen in distribution do not protect: for the MNIST prior on Fashion-MNIST all three rules are 2.1–2.8 dB below TV in PSNR and produce 0.40–0.51 false lesions per image against 0.03. Hence PSNR, detectability and fabricated structure trade against each other, and only a matched prior leaves a point at which PnP is competitive with TV on all three.

Table 9: Operating points for lesion type $\sigma_l = 0.75$, $c = 0.5$ on the test sets (one denoiser per prior; points chosen on held-out in-distribution validation images). PSNR is that of the lesion-absent reconstruction.

Prior → test set	Operating point	PSNR (dB)	pAUC	FL
MNIST → MNIST (in-dist.)	TV (PSNR-tuned)	20.11	0.910	0.005
	PnP, PSNR-tuned	24.07	0.693	0.000
	PnP, constrained	20.84	0.849	0.060
	PnP, lesion	19.57	0.916	0.315
MNIST → Fashion (shifted)	TV (PSNR-tuned)	19.37	0.914	0.030
	PnP, PSNR-tuned	16.58	0.822	0.420
	PnP, constrained	17.32	0.886	0.405
	PnP, lesion	16.78	0.903	0.505
Fashion → Fashion (in-dist.)	TV (PSNR-tuned)	19.37	0.914	0.030
	PnP, PSNR-tuned	21.58	0.751	0.005
	PnP, constrained	20.41	0.897	0.085
	PnP, lesion	19.07	0.870	0.255
Fashion → MNIST (shifted)	TV (PSNR-tuned)	20.11	0.910	0.005
	PnP, PSNR-tuned	21.67	0.759	0.000
	PnP, constrained	21.13	0.855	0.015
	PnP, lesion	21.21	0.898	0.040

4.10 E11: The matrix-free operator reproduces the dense pipeline

Table 10 compares the two implementations on the same 200 noisy test images ($\sigma = 0.05$, 20 angles, bicubic fine-grid data). The projections agree to 3×10^{-7} in relative norm, and the three reconstruction methods agree to $\leq 10^{-5}$ (Tikhonov, TV) and 3×10^{-4} (PnP) with identical PSNR to three decimals. The randomized SVD recovers the first 400 singular values to within $10^{-4}\, s_1$ and the same 465-dimensional well-determined subspace (319 weakly determined directions) as the dense SVD, and the resulting strong/weak error split agrees with the dense one to within 1% of the error norm. At this size the dense implementation is much faster for the iterative solvers (Table 10), since the adjoint is evaluated by differentiation and TV needs about 3×10^3 forward–adjoint pairs per image; PnP is only 1.2 times slower because the denoiser dominates its cost. The matrix-free operator is therefore a requirement for larger images, where the dense matrix cannot be stored, rather than an optimisation at 28×28.

4.11 E12: Poisson dose sweep

Table 11 and Figure 14 give the PnP–TV difference over the dose range (200 test images per set; the Shepp–Logan phantom is one image with one noise draw per dose, so it has no interval). The qualitative picture of Stage 1 is reproduced.

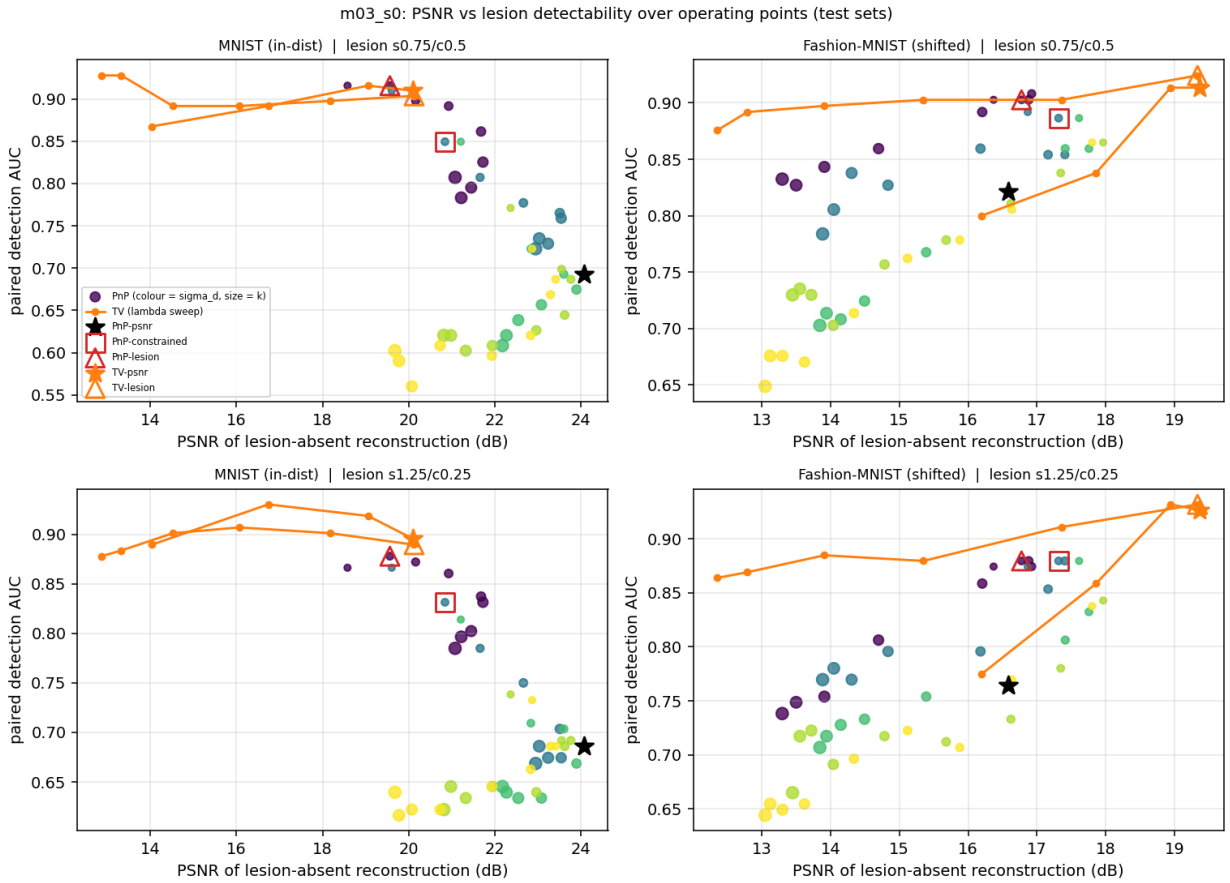


Figure 13: PSNR of the lesion-absent reconstruction versus paired detection AUC over all operating points of PnP (colour: σ_d; marker size: iterations) and TV (line: weight sweep) for the MNIST-trained denoiser. Markers show the selected operating points. Left: in distribution; right: Fashion-MNIST (shifted).

Table 10: Dense versus matrix-free pipeline (200 images, GPU). Rel. diff.: relative difference between the two reconstructions. Peak GPU memory for the whole comparison: 184 MB.

Method	Rel. diff.	PSNR (dense = MF)	Time dense (s)	Time MF (s)
Tikhonov	4.6×10^{-6}	16.36	0.08	0.26
TV-ADMM	9.7×10^{-6}	19.01	0.16	7.50
PnP ($k = 25$)	2.8×10^{-4}	23.10	0.76	0.94

Prior-matched data. On MNIST the learned prior beats TV at every dose, by +3.7 dB at $I_0 = 100$–300 (effective $\sigma \approx 0.03$–0.06), against +3.84 dB at Gaussian $\sigma = 0.05$, and for 99–100% of the images at $I_0 \geq 100$. The advantage shrinks at both ends of the dose range, to +0.8 dB at $I_0 = 10$, in line with the Gaussian result at $\sigma = 0.2$ (+0.97 dB). EMNIST keeps a smaller advantage (+0.6 to +1.8 dB).

Shifted data. The MNIST-trained prior is below TV on Fashion-MNIST (−2.0 to −3.1 dB; PnP wins on only 6–14% of the images) and on the phantom (−4.4 to −6.1 dB), as in Stage 1 (−3.10 and −4.33 dB). The share of the error in the complement of the well-determined subspace is 0.68–0.99 of TV's on MNIST but 1.04–1.19 on Fashion-MNIST and 1.73–1.79 on the phantom (for $I_0 \leq 300$), so the mismatch error again sits partly in the weakly determined directions. These ratios are less favourable to PnP than the Stage 1 values and are not directly comparable, since the noise model, the tuned settings and the subspace definition differ.

Weighted data term. Using the Poisson weights in PnP gave no consistent benefit: the weighted variant was 0.14–0.59 dB below the unweighted one in 13 of the 15 non-phantom cells, and above it in two (EMNIST and Fashion-MNIST at $I_0 = 300$), where the tuned (σ_d, k) differed. We therefore report the unweighted scheme.

Table 11: PnP−TV PSNR difference (dB, mean ± standard error of the paired difference over 200 images) by photon count I_0, with the corresponding effective sinogram noise standard deviation; and the ratio of PnP's to TV's reconstruction error in the complement of the well-determined subspace (in brackets). Phantom: one image.

I_0 (photons/ray) effective σ	1000 0.018	300 0.032	100 0.056	30 0.106	10 0.198
MNIST (train dist.)	+2.87 ± .07 (0.70)	+3.68 ± .08 (0.68)	+3.65 ± .08 (0.71)	+2.39 ± .12 (0.88)	+0.82 ± .12 (0.99)
EMNIST	+1.40 ± .09 (0.75)	+1.59 ± .08 (0.83)	+1.82 ± .08 (0.86)	+0.77 ± .11 (1.02)	+0.62 ± .10 (1.00)
Fashion-MNIST	−2.45 ± .13 (1.04)	−3.05 ± .16 (1.18)	−2.74 ± .16 (1.18)	−3.01 ± .14 (1.19)	−1.98 ± .13 (1.07)
Shepp–Logan	−4.41 (0.90)	−6.05 (1.79)	−5.05 (1.73)	−5.49 (1.77)	−4.75 (1.75)

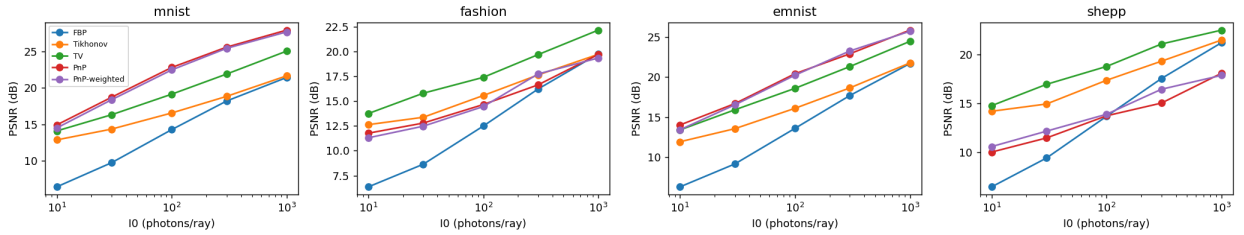


Figure 14: PSNR versus photon count for FBP, Tikhonov, TV, PnP and the Poisson-weighted PnP on the four test sets (MNIST-trained denoiser, hyperparameters tuned on MNIST validation images at each dose).

4.12 E13: Lesion insertion under Poisson noise

We repeated the lesion experiment (Section 3.8; same lesions, observer, locations and validity filter; three denoisers per prior, 200 images per set) with Poisson data at $I_0 = 100$ and $I_0 = 30$ and the matrix-free operator, tuning each method on its prior's own validation distribution (Table 12, Figure 15).

Erasure persists. The paired AUC of PnP is below TV's in all four prior/test-set combinations and at both doses (differences of -0.01 to -0.28 at $I_0 = 100$, with one tie at 1.0 for the largest high-contrast lesion, and -0.02 to -0.19 at $I_0 = 30$). For the MNIST prior on MNIST ($\sigma_l = 0.75$, $c = 0.5$) the paired AUC is 0.60 against 0.89 for TV and the retention is 0.29 against 0.60 at $I_0 = 100$, which reproduces the Gaussian-noise values of Stage 1 (0.69 vs. 0.91; 0.28 vs. 0.59), although PnP is 3.7 dB better in PSNR on the same images. At $I_0 = 30$ the gap narrows (0.63 vs. 0.77) because TV degrades faster.

Invention under mismatch persists. For the MNIST-trained prior on Fashion-MNIST, PnP produces more false lesions than TV for lesions with $\sigma_l \geq 0.75$ in 11 of the 12 dose/lesion-type cells, by factors of 1.7–5.5 (0.25–0.70 against 0.06–0.28 per image at $I_0 = 100$; the exception is 0.25 against 0.28). The matched Fashion-trained prior, on its own data and on MNIST, stays at ≤ 0.09 false lesions per image. Retention above that of TV for the mismatched prior should not be read as fidelity, since invented contrast contributes to it.

New at the lower dose: in-distribution invention. For the MNIST prior on MNIST at $I_0 = 30$, PnP produces 0.058–0.073 false lesions per image for the four lesion types with $0.75 \leq \sigma_l \leq 1.25$, against 0.000–0.035 for TV, whereas at $I_0 = 100$ it produces at most 0.01 for any type. At high noise the in-distribution prior therefore starts to create lesion-like structure as well as to erase it, although at a rate far below that under mismatch.

TV has a failure of its own. For the smallest low-contrast lesion at $I_0 = 100$ the TV reconstruction contains 0.63 false lesions per image (noise blobs), against 0.01 for PnP. The statement that PnP creates more false lesions than TV under mismatch therefore applies to lesions of $\sigma_l \geq 0.75$ px, and under a matched prior is not supported at all.

Uncertainty is limited by the design: the error bars in Figure 15 show the variation over three denoisers only, TV is a single run, and with about 170 valid lesions per cell the sampling error of a paired AUC is of the order of ± 0.03, so differences below that, such as the ties for the largest lesions, do not separate the methods. A tercile analysis of PnP's paired AUC by lesion observability showed a decrease with observability on MNIST images and no trend on Fashion-MNIST images; since it depends on the image set it is likely to reflect the background around the lesion rather than the operator, and we do not interpret it.

Table 12: Lesion insertion under Poisson noise, mean over three denoisers per prior (200 test images; 20 angles; bicubic data), for two of the eight lesion types. pAUC: paired AUC; Ret.: retention; FL: false lesions per lesion-absent image. Each method is tuned on its own prior's validation distribution at the given dose.

		$I_0 = 100$						$I_0 = 30$					
		pAUC		Ret.		FL		pAUC		Ret.		FL	
Prior → test set	σ_l/c	TV	PnP	TV	PnP	TV	PnP	TV	PnP	TV	PnP	TV	PnP
MNIST → MNIST	0.75/0.5	.89	.60	.60	.29	.010	.007	.77	.63	.48	.34	.005	.067
	1.25/0.25	.88	.60	.60	.27	.020	.007	.75	.65	.46	.46	.005	.068
MNIST → Fashion	0.75/0.5	.81	.72	.64	.84	.240	.443	.73	.63	.42	.57	.230	.518
	1.25/0.25	.84	.72	.65	.84	.255	.438	.71	.61	.42	.54	.225	.518
Fashion → Fashion	0.75/0.5	.79	.64	.46	.23	.075	.020	.69	.58	.25	.11	.020	.038
	1.25/0.25	.85	.71	.48	.25	.075	.020	.68	.56	.27	.16	.040	.038
Fashion → MNIST	0.75/0.5	.88	.66	.48	.31	.000	.003	.78	.64	.33	.27	.000	.007
	1.25/0.25	.89	.66	.49	.33	.000	.003	.78	.65	.34	.34	.000	.007



Figure 15: Paired detection AUC (top rows) and false lesions per lesion-absent image (bottom rows) under Poisson noise, $I_0 = 100$ (upper panel) and $I_0 = 30$ (lower panel). Bars: mean over three denoisers; error bars: s.d. over seeds.

4.13 E14: Data, noise model and simulator

The geometry fit, the noise-model checks and the dose thinning are described in Sections 3.11–3.12. The decisive check for everything that follows is the simulator: Table 13 compares the TV PSNR obtained from the real (thinned) observations with that obtained from our crime-free simulation of the same slices. The agreement is within 0.15 dB at all conditions; for all 40 slices at $K = 60$ the values are 31.91 vs. 32.04 dB (full dose) and 29.82 vs. 29.83 dB (quarter dose), and for FBP-Hann 30.65 vs. 30.59 and 28.73 vs. 28.66 dB. The nine-condition check uses the first 8 test slices only. Residual differences are small compared with the method differences studied below (≤ 2.3 dB), but they are not zero, and the simulated data may be somewhat less noisy than the real ones at high counts (dispersion index above).

Table 13: TV PSNR (dB) from real observations (thinned to the dose, subsampled to K angles) and from the crime-free simulation of the same slices (first 8 test slices).

K	30			60			120		
dose	1	0.25	0.1	1	0.25	0.1	1	0.25	0.1
real	29.34	27.95	26.57	30.15	28.40	26.73	30.61	28.54	27.89
simulated	29.41	27.90	26.52	30.27	28.45	26.72	30.76	28.55	27.85

Table 14: Tuned parameters (20 validation slices per condition): Hann cut-off, TV weight (the TV-long weight was the same in every condition), and (σ_d, k) of PnP-const and PnP-TVinit.

K	dose	Hann	TV λ	PnP-const (σ_d, k)	PnP-TVinit (σ_d, k)
30	1	0.2	0.003	(0.02, 12)	(0.02, 4)
30	0.25	0.16	0.01	(0.05, 8)	(0.08, 4)
30	0.1	0.12	0.03	(0.08, 12)	(0.05, 8)
60	1	0.3	0.003	(0.02, 8)	(0.02, 2)
60	0.25	0.2	0.01	(0.08, 4)	(0.02, 8)
60	0.1	0.12	0.03	(0.05, 12)	(0.02, 12)
120	1	0.4	0.003	(0.05, 2)	(0.0039, 8)†
120	0.25	0.2	0.01	(0.02, 12)	(0.0062, 8)
120	0.1	0.16	0.01	(0.05, 8)	(0.128, 1)†

† optimum on the boundary of the grid after the automatic extensions.

4.14 E15: Learned prior versus TV on real slices at $K = 60$

Table 15 compares PnP variants with TV on the 40 test slices at $K = 60$ (denoiser of depth 8 and a deeper one, 17 layers, trained for 20 000 steps). With the re-tuned starting image, PnP-const is slightly but significantly below TV (-0.17 and -0.10 dB) and PnP-TVinit is at TV level at full dose ($+0.03$ dB, interval $[-0.00, +0.06]$) and $+0.21$ dB above it at quarter dose. The deeper denoiser does not help (PnP-const -0.30 and -0.25 dB). Annealing σ_d over the iterations is worse than constant σ_d at full dose (-0.65 dB) and better at quarter dose ($+0.12$ dB, but not for the deeper denoiser: $+0.05$, interval including zero). One denoiser pass on the TV image gives TV (within ± 0.05 dB at the depth-8 network). In short, on clinical-size slices and at $K = 60$ the learned prior is competitive with TV in PSNR rather than ≈ 4 dB better as in the 28×28 experiments, with the largest gain, 0.2 dB, from combining the two.

4.15 E16: Accuracy over the dose $\times$ angle grid

Table 16 gives the mean PSNR and SSIM of the five methods in the nine conditions of the lesion-free arm, and Table 17 the paired differences. Three statements hold in all nine conditions: PnP-const and TV-long beat FBP-Hann in PSNR (PnP-const by 0.93–1.36 dB, TV-long by 0.81–2.28 dB) and in SSIM, with all intervals excluding zero. PnP-const is *not* better than TV-long on average: its PSNR difference lies between -0.93 and $+0.33$ dB, significantly positive in three conditions ($K = 30$ and 60 at dose 0.1, and $K = 120$ at full dose), significantly negative in four ($K = 30$, 60 at full dose; $K = 60$ at quarter dose; $K = 120$ at dose 0.1) and not significant in two; its SSIM is below TV-long's in eight of nine conditions, significantly so in seven, and above it in one ($+0.001$, not significant). The sign does not follow dose or angle count monotonically (the low-dose, few-view advantage at $K = 30$ and 60 is reversed at $K = 120$, dose 0.1). PnP-TVinit has the highest mean

Table 15: $K = 60$, 40 test slices: PSNR (dB) and SSIM, and the paired PSNR difference to TV (mean, 95% bootstrap interval over patients; * excludes 0). d8: 8-layer denoiser; d17: 17-layer denoiser.

	dose 1			dose 0.25		
Method	PSNR	SSIM	−TV	PSNR	SSIM	−TV
FBP-Hann	30.65	0.738	-1.27^*	28.73	0.699	-1.09^*
TV	31.91	0.786	–	29.82	0.750	–
PnP-const [d8]	31.74	0.773	-0.17^* $[-0.21, -0.13]$	29.72	0.734	-0.10^* $[-0.18, -0.03]$
PnP-TVinit [d8]	31.94	0.784	$+0.03$ $[-0.00, +0.06]$	30.03	0.753	$+0.21^*$ $[+0.17, +0.25]$
PnP-anneal [d8]	31.26	0.769	-0.65^* $[-0.77, -0.52]$	29.94	0.750	$+0.12^*$ $[+0.05, +0.18]$
TV+denoise [d8]	31.91	0.785	$+0.00$ $[-0.02, +0.03]$	29.77	0.747	-0.05 $[-0.10, +0.01]$
PnP-const [d17]	31.61	0.777	-0.30^* $[-0.35, -0.26]$	29.57	0.737	-0.25^* $[-0.31, -0.19]$
PnP-TVinit [d17]	31.89	0.785	-0.02 $[-0.06, +0.03]$	29.97	0.755	$+0.15^*$ $[+0.12, +0.18]$
PnP-anneal [d17]	31.20	0.772	-0.71^* $[-0.85, -0.58]$	29.87	0.753	$+0.05$ $[-0.01, +0.12]$
TV+denoise [d17]	31.90	0.786	-0.02 $[-0.05, +0.02]$	29.84	0.752	$+0.02^*$ $[+0.00, +0.04]$

PSNR in seven of nine conditions, by at most 0.49 dB over TV-long (significantly higher in five, lower in two: $K = 30$, full dose, -0.15 dB, and $K = 60$, full dose, -0.03 dB). Two of its cells are not denoiser effects (Section 3.13). TV-long is slightly worse than TV at $K = 120$, full dose (32.50 vs. 32.64 dB), which shows that the early-stopped 150-iteration TV is itself a tuned object and that the weight grid is coarse.

4.16 E17: Lesion detection

Grid (Table 18). Pooled over the two lesion sizes (contrast 0.06; up to 160 lesions per size), PnP-const has a higher paired AUC than TV-long in all nine conditions, by 0.03 to 0.16, with intervals excluding zero in six (all three $K = 120$ conditions, $K = 60$ at doses 0.25 and 0.1, $K = 30$ at dose 0.1); it is not significant at full dose for $K = 30$ and 60. Against FBP-Hann, however, PnP-const is never significantly better (three positive, six negative or zero differences) and is significantly worse in two conditions ($K = 30$, full dose, -0.075; $K = 120$, quarter dose, -0.059): the learned prior removes the detection deficit of TV rather than adding to FBP's detectability. TV-long, tuned for PSNR, has a lower AUC than FBP-Hann in all nine conditions (significantly in seven, down to -0.20 at $K = 60$, dose 0.1) while being 0.81–2.28 dB better in PSNR: PSNR and detectability rank the methods in opposite orders. PnP-TVinit exceeds TV-long in AUC in all nine conditions (significantly in seven; +0.02 to +0.17).

Retention. At $K = 60$ and full dose the regularised methods retain only 7–32% of the local-contrast difference of the inserted lesion (FBP-Hann 31–67%); the detection scores of TV and PnP are therefore obtained mainly from noise suppression, not from preserving lesion contrast.

Earlier analysis at $K = 60$ (E17b). With four lesion types (two sizes, contrasts 0.03 and 0.06; 160 lesions per type pooled), re-tuned on 20 slices, the pooled AUC difference PnP-const − TV-long is +0.023 ($[-0.013, +0.061]$) at full dose and +0.073 ($[+0.023, +0.128]$) at quarter dose, in line with the grid values (+0.050 and +0.100; different lesion sets and noise). At both doses PnP-const is level with FBP-Hann (-0.008, +0.016; not significant) and TV is below it (-0.037, not significant; -0.056, $[-0.11, -0.00]$). At $\sigma_l = 4$ px and $c = 0.06$ all methods reach AUC 0.95–0.96 at full dose, so the large lesions do not discriminate between methods at that dose (ceiling).

Table 16: Mean PSNR (dB, upper block) and SSIM (lower block) of the lesion-free reconstructions, 40 patients (best in each condition in bold). Columns: K angles / dose.

	$K = 30$			$K = 60$			$K = 120$		
Method	1	0.25	0.1	1	0.25	0.1	1	0.25	0.1
PSNR (dB)									
FBP-Hann	29.02	27.72	26.63	30.59	28.66	27.11	31.70	29.18	27.94
TV	30.98	29.15	27.69	32.04	29.83	27.93	32.64	30.07	29.13
TV-long	**31.31**	29.21	27.72	**32.12**	29.88	27.98	32.50	30.11	29.21
PnP-const	30.38	29.05	27.86	31.85	29.68	28.31	32.64	30.23	28.86
PnP-TVinit	31.15	**29.22**	**27.96**	32.09	**30.04**	**28.47**	**32.92**	**30.46**	**29.26**
SSIM									
FBP-Hann	0.692	0.666	0.655	0.738	0.699	0.680	0.763	0.725	0.693
TV	0.765	0.733	0.710	0.788	0.750	0.717	0.799	0.756	0.741
TV-long	**0.773**	**0.737**	0.713	**0.790**	0.753	0.720	0.797	0.758	**0.745**
PnP-const	0.745	0.726	0.707	0.775	0.733	0.721	0.793	0.749	0.731
PnP-TVinit	0.767	0.730	**0.721**	0.786	**0.753**	**0.731**	**0.805**	**0.761**	0.740

Table 17: Paired differences over patients (mean, 95% bootstrap interval; * excludes 0), 40 patients.

K	dose	PSNR: PnP-const − TV-long	SSIM: PnP-const − TV-long	PSNR: PnP-TVinit − TV-long	PSNR: PnP-const − FBP-Hann
30	1	-0.93^* [−1.09, −0.75]	-0.027^* [−0.035, −0.021]	-0.15^* [−0.24, −0.06]	$+1.36^*$ [+1.15, +1.58]
30	0.25	−0.15 [−0.36, +0.08]	-0.011^* [−0.019, −0.002]	+0.01 [−0.19, +0.24]	$+1.33^*$ [+1.06, +1.62]
30	0.1	$+0.13^*$ [+0.03, +0.23]	−0.006 [−0.014, +0.001]	$+0.24^*$ [+0.19, +0.29]	$+1.22^*$ [+0.98, +1.48]
60	1	-0.28^* [−0.33, −0.22]	-0.015^* [−0.019, −0.011]	-0.03^* [−0.06, −0.00]	$+1.25^*$ [+0.97, +1.56]
60	0.25	-0.20^* [−0.28, −0.13]	-0.020^* [−0.027, −0.012]	$+0.16^*$ [+0.11, +0.21]	$+1.02^*$ [+0.65, +1.41]
60	0.1	$+0.33^*$ [+0.23, +0.43]	+0.001 [−0.003, +0.005]	$+0.49^*$ [+0.37, +0.62]	$+1.20^*$ [+1.04, +1.37]
120	1	$+0.14^*$ [+0.05, +0.24]	-0.004^* [−0.007, −0.001]	$+0.41^*$ [+0.32, +0.50]	$+0.95^*$ [+0.54, +1.39]
120	0.25	+0.13 [−0.02, +0.27]	-0.009^* [−0.013, −0.006]	$+0.35^*$ [+0.25, +0.45]	$+1.05^*$ [+0.89, +1.22]
120	0.1	-0.34^* [−0.47, −0.22]	-0.013^* [−0.020, −0.007]	+0.06 [−0.07, +0.19]	$+0.92^*$ [+0.73, +1.13]

Table 18: Paired AUC differences ($\sigma_l \in \{2, 4\}$ px pooled, $c = 0.06$, 40 patients; mean, 95% interval; * excludes 0).

K	dose	PnP-const − TV-long	PnP-const − TV	PnP-const − FBP-Hann	TV-long − FBP-Hann	PnP-TVinit − TV-long
30	1	+0.044 [−0.009, +0.097]	+0.006 [−0.050, +0.059]	-0.075^* [−0.116, −0.037]	-0.119^* [−0.169, −0.072]	$+0.066^*$ [+0.019, +0.113]
30	0.25	+0.034 [−0.025, +0.094]	$+0.078^*$ [+0.013, +0.144]	−0.006 [−0.056, +0.041]	−0.041 [−0.103, +0.025]	$+0.069^*$ [+0.006, +0.131]
30	0.1	$+0.091^*$ [+0.019, +0.162]	$+0.137^*$ [+0.059, +0.216]	+0.031 [−0.028, +0.091]	−0.059 [−0.141, +0.019]	+0.053 [−0.022, +0.128]
60	1	+0.050 [−0.006, +0.103]	+0.000 [−0.050, +0.050]	−0.037 [−0.091, +0.012]	-0.087^* [−0.144, −0.031]	$+0.069^*$ [+0.031, +0.109]
60	0.25	$+0.100^*$ [+0.050, +0.150]	$+0.078^*$ [+0.028, +0.131]	−0.009 [−0.059, +0.041]	-0.109^* [−0.175, −0.047]	$+0.084^*$ [+0.031, +0.141]
60	0.1	$+0.163^*$ [+0.078, +0.247]	$+0.144^*$ [+0.072, +0.219]	−0.041 [−0.097, +0.009]	-0.203^* [−0.291, −0.119]	$+0.156^*$ [+0.084, +0.231]
120	1	$+0.109^*$ [+0.066, +0.156]	$+0.084^*$ [+0.044, +0.125]	+0.003 [−0.031, +0.041]	-0.106^* [−0.162, −0.053]	$+0.091^*$ [+0.050, +0.137]
120	0.25	$+0.112^*$ [+0.041, +0.184]	$+0.081^*$ [+0.006, +0.156]	-0.059^* [−0.103, −0.013]	-0.172^* [−0.247, −0.094]	$+0.166^*$ [+0.109, +0.222]
120	0.1	$+0.084^*$ [+0.025, +0.147]	$+0.084^*$ [+0.028, +0.147]	+0.013 [−0.047, +0.072]	-0.072^* [−0.128, −0.016]	+0.022 [−0.041, +0.081]

4.17 E18: False structures on real slices

Table 21 (CT rows) gives the mean number of false blobs per lesion-free slice at one times the nominal lesion strength for the four conditions that are also used in the shift study. On real slices all regularised methods make few such blobs: PnP-const 0.24–1.38 per slice, TV-long 0.00–6.06, FBP-Hann 5.8–26.8. Over the nine conditions the paired difference PnP-const − TV-long at $m = 1$ (Table 19) is significantly positive in three ($K = 60$ full dose: $+0.46$; $K = 120$ full and quarter dose: $+0.24$, $+1.35$), significantly negative in three ($K = 30$ quarter dose: -5.31; $K = 60$ quarter dose: -0.40; $K = 120$ dose 0.1: -0.93) and not significant in three, with no pattern in dose or angle count. At $m = 1.5$ the excess of PnP-const over TV-long is at most 0.45 blobs per slice (K=120, quarter dose). In the $K = 60$ experiment (E17b, threshold $m = 0.5$) PnP-const has fewer false blobs than TV at both doses (-1.7 at full dose, -6.6 at quarter dose). The overlays of the worst false blobs (`step4b_gallery_*.png`) show that the largest ones occur in FBP-Hann and TV as well, so they are produced by the data (few views, noise, streaks); only a few look specific to the prior, for example a crescent in one slice at full dose and streaks in the PnP-TVinit image of one slice at quarter dose. We therefore find *no evidence that the prior creates many lesion-like structures in distribution*, but we cannot exclude rare ones: 40 slices and one noise draw per condition do not bound a rare event.

Table 19: Paired differences in false blobs per lesion-free slice on real CT slices (40 patients; mean, 95% interval; * excludes 0), pooled over $\sigma_l \in \{2, 4\}$; m: threshold in units of the nominal lesion response.

K	dose	$m = 1$: PnP-const − TV-long	$m = 1$: PnP-const − TV	$m = 1.5$: PnP-const − TV-long
30	1	-1.19 $[-2.79, +0.05]$	-0.54 $[-1.57, +0.36]$	-0.23 $[-0.64, +0.11]$
30	0.25	-5.31^* $[-9.15, -2.31]$	-4.81^* $[-8.41, -2.00]$	-2.42^* $[-4.56, -0.82]$
30	0.1	$+0.21$ $[-0.20, +0.64]$	$+0.31$ $[-0.12, +0.76]$	$+0.00$ $[-0.14, +0.14]$
60	1	$+0.46^*$ $[+0.12, +0.88]$	$+0.19^*$ $[+0.01, +0.40]$	$+0.12^*$ $[+0.03, +0.25]$
60	0.25	-0.40^* $[-0.68, -0.17]$	-0.60^* $[-1.11, -0.19]$	-0.11^* $[-0.25, -0.01]$
60	0.1	-0.07 $[-0.25, +0.07]$	-0.05 $[-0.20, +0.09]$	-0.01 $[-0.06, +0.04]$
120	1	$+0.24^*$ $[+0.04, +0.51]$	$+0.11^*$ $[+0.01, +0.25]$	$+0.06$ $[+0.00, +0.15]$
120	0.25	$+1.35^*$ $[+0.53, +2.30]$	$+1.35^*$ $[+0.54, +2.29]$	$+0.45^*$ $[+0.16, +0.81]$
120	0.1	-0.93^* $[-1.50, -0.45]$	-0.88^* $[-1.48, -0.39]$	-0.11^* $[-0.26, -0.01]$

4.18 E19: Distribution shift to non-anatomical phantoms

The CT-trained prior and all tuned parameters are applied, without retuning, to 20 random piecewise-constant phantoms (ellipses, rectangles and a stripe or checker block on a faint disc, $\mu/\mu_{\max}$ scale; nothing like a CT slice, but the kind of image TV favours; same phantoms in every condition) with the same lesion protocol. The shift was first run at $K = 60$ (doses 1 and 0.25) and then extended to $K = 30$ and 120; the $K = 60$ analysis used four lesion types, the others two, so the designs differ slightly.

Accuracy and detection (Table 20). On the phantoms PnP-const is below TV-long in PSNR in all six conditions (-0.11 to -0.75 dB; significantly in three of the four conditions for which an interval was computed) and in SSIM in all four (-0.095 to -0.190, all significant, against -0.004 to -0.027 on CT). At quarter dose its SSIM is also below FBP-Hann's on the phantoms (-0.045 at $K = 30$ and -0.041 at $K = 120$, both significant). The detection advantage over TV-long is significant in one of six shift conditions ($K = 120$, full dose: $+0.075$, $[+0.013, +0.137]$), against

three of the six corresponding CT conditions ($K = 60$ quarter dose, $K = 120$ at both doses); the shift effect is significantly negative only at $K = 120$, quarter dose (-0.175, $[-0.297, -0.056]$). The CT advantage is therefore not reliably transferred to the shifted images, but at 20 phantoms the interval widths of 0.12–0.19 do not allow us to say that it is lost either.

False blobs (Tables 20–21). The sign of PnP-const − TV at 1× lesion strength follows the number of angles: negative (or not different from zero) at $K = 30$ and negative at full-dose $K = 60$, mixed at $K = 60$ quarter dose ($\sigma_l = 2$: -7.5; $\sigma_l = 4$: $+9.1$), and positive at $K = 120$ ($+9.0$ to $+13.0$ per phantom at both doses; at 1.5× against TV-long $+10.2$ at full dose and $+3.8$ at quarter dose, both significant). The absolute counts show why (Table 21): the count of TV-long varies from 3.2 to 83.9 per phantom over the four conditions, that of PnP-const only from 12.8 to 50.3; at $K = 30$ TV-long makes two to three times as many false blobs as FBP-Hann (30.2 vs. 15.5 at quarter dose, 83.9 vs. 31.6 at full dose), at $K = 120$ a third or less (3.2 vs. 8.7; 32.1 vs. 118.7). So the "excess" of PnP-const at $K = 120$ is relative to a TV that is unusually clean there. Relative to *FBP-Hann* the picture is more informative: at full dose PnP-const has fewer false blobs on the phantoms (-7.1 at $K = 30$, -68.4 at $K = 120$, both significant), but at quarter dose it has more at both K ($+1.0$, not significant, and $+4.0$ $[+2.4, +5.9]$ at 1×; $+2.2$ $[+1.1, +3.4]$ and $+2.7$ $[+1.4, +4.4]$ at 1.5×), in contrast to real slices, where it has significantly fewer than FBP-Hann in both quarter-dose conditions (-9.0 and -4.4). In this sense the shift turns the prior from suppressing to adding false blobs relative to the unregularised reconstruction at quarter dose, by a few blobs per phantom. The pooled counts hide a dependence on lesion size: at 1×, for $\sigma_l = 2$ PnP-const has more false blobs than FBP-Hann on the phantoms in three of the four conditions ($+10.1$, $+9.7$ and $+10.6$ at $K = 30$ full and quarter dose and $K = 120$ quarter dose, all significant), for $\sigma_l = 4$ fewer in all four (-24.3, -7.7, -30.9, -2.5). At $K = 120$, full dose, PnP-TVinit with almost no denoising ($\sigma_d = 0.0039$) also produces 14.0 more blobs than TV-long ($[+7.4, +22.0]$ at 1×, for all σ_l), so at least part of the excess at $K = 120$ is not due to the denoiser; we did not isolate the cause.

Table 20: Distribution shift to phantoms (20 phantoms; mean, 95% interval; * excludes 0). Left: false blobs per phantom at 1× lesion strength, PnP-const − TV, by lesion size. Right: PnP-const − TV-long on the phantoms: pooled paired AUC, PSNR (dB) and SSIM. $K = 60$ rows: AUC from four lesion types (E17b), PSNR as a difference of means without an interval, SSIM not computed.

		false blobs, 1×				
K	dose	$\sigma_l = 2$	$\sigma_l = 4$	AUC	PSNR	SSIM
30	1	-70.7^* $[-107.0, -38.7]$	-31.8^* $[-48.7, -16.8]$	-0.037 $[-0.113, +0.037]$	-0.11 $[-0.37, +0.18]$	-0.095^* $[-0.125, -0.063]$
30	0.25	-21.1^* $[-34.3, -10.3]$	-1.8 $[-5.8, +1.8]$	$+0.056$ $[-0.031, +0.144]$	-0.53^* $[-0.64, -0.40]$	-0.190^* $[-0.204, -0.172]$
60	1	-20.1^* $[-34.4, -7.6]$	-6.5^* $[-12.6, -1.4]$	-0.019 $[-0.06, +0.02]$	-0.30 (no interval)	n/a
60	0.25	-7.5^* $[-12.9, -2.8]$	$+9.1^*$ $[+6.0, +12.8]$	$+0.050$ $[-0.01, +0.11]$	-0.75 (no interval)	n/a
120	1	$+13.0^*$ $[+8.8, +17.5]$	$+9.2^*$ $[+5.4, +13.2]$	$+0.075^*$ $[+0.013, +0.137]$	-0.51^* $[-0.56, -0.45]$	-0.179^* $[-0.190, -0.168]$
120	0.25	$+10.4^*$ $[+5.9, +16.0]$	$+9.0^*$ $[+5.9, +12.3]$	-0.062 $[-0.162, +0.031]$	-0.62^* $[-0.68, -0.57]$	-0.155^* $[-0.162, -0.149]$

5 Discussion

Ten findings emerge from this study.

First, the learned prior's accuracy advantage over TV is conditional rather than universal. It is large and significant at moderate noise and for moderate undersampling ($\approx +3.9$ dB at $\sigma = 0.05$ with 20–40 angles), smaller at the extremes of both ranges, and in the near noise-free regime TV remains ahead by an amount that depends on how well the data model matches the reconstruction

Table 21: Mean false blobs per lesion-free image at 1× nominal lesion strength (mean of $\sigma_l = 2$ and 4) on real CT slices and on the phantoms, and PnP-const − FBP-Hann (mean, 95% interval over patients or phantoms).

set	K, dose	FBP-Hann	TV	TV-long	PnP-const	PnP-TVinit	PnP-const − FBP
CT	30, 0.25	9.78	5.56	6.06	0.75	0.64	-9.03^* $[-13.21, -5.20]$
CT	30, 1	5.78	1.04	1.69	0.50	0.32	-5.28^* $[-7.54, -3.29]$
CT	120, 0.25	5.79	0.02	0.02	1.38	0.39	-4.41^* $[-7.34, -2.00]$
CT	120, 1	26.79	0.12	0.00	0.24	0.18	-26.55^* $[-43.93, -12.63]$
phantom	30, 0.25	15.48	27.95	30.20	16.50	11.65	$+1.03$ $[-0.68, +2.75]$
phantom	30, 1	31.60	75.72	83.90	24.48	57.30	-7.13^* $[-10.68, -4.02]$
phantom	120, 0.25	8.72	3.08	3.18	12.75	11.48	$+4.03^*$ $[+2.35, +5.90]$
phantom	120, 1	118.68	39.20	32.08	50.30	46.12	-68.38^* $[-92.68, -45.23]$

model (Section 4.5). Reporting a single aggregate accuracy number for a learned reconstruction method without specifying the operating regime and the data model is therefore misleading.

Second, PnP-PGD with an unconstrained CNN denoiser does not behave like a convergent optimization scheme: accuracy peaks after a handful of iterations and then degrades while the data residual rises (Section 4.2). This is consistent with the observation that convergence guarantees need constraints on the denoiser (e.g. non-expansiveness) [11]. Our decomposition adds a mechanism: with step $\eta = 1/L$ the data step contracts the error in direction i by $(1 - (s_i/s_1)^2)^K$, so after the few iterations selected by validation, directions with $s_i < 0.3\, s_1$ are barely corrected and the reconstruction there is largely the denoiser's. Iteration count is thus an implicit regularization hyperparameter, and not an implementation detail.

Third, the reversal under distribution shift is a failure of the prior, not of the scheme. The matched-prior control (Section 4.7) shows that PnP beats TV on Fashion-MNIST when the denoiser was trained on it, and the error decomposition shows that the mismatched prior places up to three times the true energy in the weakly determined subspace. The effect is robust: it exceeds seed-to-seed variation by an order of magnitude, persists under every data model we tested, and cannot be removed by changing σ_d or the iteration count. At the same time the failure is not confined to the weakly determined subspace, so "hallucination" in the null-space sense is an incomplete description.

Fourth, PSNR is not task performance. In distribution, the PSNR-tuned PnP reconstruction is about 4 dB better than TV yet detects small inserted lesions markedly less reliably, because it attenuates them (Section 4.8). The errors take two forms: erasure of unfamiliar small features when the prior is adequate, and invention of lesion-like blobs when it is not. Re-tuning trades one against the other (Section 4.9); only a matched prior leaves a point that is competitive with TV on PSNR, detection and false structure at once, and operating points chosen in distribution do not protect under shift.

Fifth, the learned method loses its advantage faster than TV under forward-model mismatch: it remains more accurate throughout the tested range but its margin falls from +3.85 dB to +1.48 dB at a 4° offset (Section 4.3). This experiment was run with data generated by the nominal operator and should be repeated on finer-grid data.

Sixth, evaluation protocol matters. The inverse crime changes the low-noise PnP–TV gap by up to 5.5 dB; a single denoiser instance would have reported anywhere between −2.83 and −3.43 dB for Fashion-MNIST depending on the seed; and operating points for the lesion task were therefore chosen on held-out images rather than on the evaluation images, to avoid overstating the benefit of re-tuning. These results are consistent with broader findings on instabilities in learned

reconstruction [1, 6] and support the view that in-distribution accuracy alone is an insufficient basis for adopting a learned prior; mismatch, shift, seed and task-based evaluation should be a standard part of such studies, not an optional addendum.

Seventh, the conclusions do not depend on the two simplifications of the early experiments. Replacing the dense matrix by a matrix-free projector with an exact adjoint changes no reconstruction beyond 3×10^{-4}, and replacing additive Gaussian noise by Poisson noise leaves the picture unchanged: the learned prior's gain over TV peaks at the same effective noise level and is lost under shift, the lesion deficit of the PSNR-tuned prior in distribution is the same, and invention under mismatch is the same. The one new effect, a small rate of in-distribution false lesions at high noise, is a reminder that the failure mode of a learned prior moves with the noise level, so conclusions about hallucination should be stated for an operating regime. The weighted Poisson data term did not help, so the Gaussian approximation is adequate in this setting.

Eighth, the advantage of the learned prior in PSNR does not carry over from the 28×28 benchmark to clinical-size CT. The +3.9 dB over TV found with a matched prior on MNIST becomes −0.93 to +0.33 dB over a TV tuned with twice the iterations on 362×362 slices of 40 test patients, and the SSIM is lower in seven of nine conditions (Section 4.15). Both iterative methods improve on FBP-Hann by 0.8–2.3 dB, so the gain over FBP comes from regularisation in general, not from the learned prior in particular, and a deeper denoiser did not change this at $K = 60$ (Section 4.14). We did not test why the small-scale and large-scale results differ; candidate explanations are the more varied anatomy (for which TV is a weaker model than for digits), a denoiser trained on only 8 960 of the 35 820 slices, the small number of PnP iterations, and a TV baseline that was tuned and given twice the iterations.

Ninth, PSNR and detectability disagree, now in a different way from the small-scale results. In Stage 1 the PSNR-tuned PnP reconstruction detected lesions worse than TV; on CT it detects them better than TV in all nine conditions (significantly in six) but no better than unregularised FBP, and the PSNR-tuned TV is worse than FBP at detection in all nine conditions (significantly in seven) although it is 0.8–2.3 dB better in PSNR (Section 4.16). A reconstruction method chosen on PSNR can therefore be the worse choice for a detection task, whichever method it is, and a comparison of learned and classical priors that reports only PSNR or SSIM would have ranked TV first and missed that the ranking of detectability is reversed. The lesions are small ($\sigma_l = 2$–4 px) and of high contrast in the clinical sense; the conclusion for clinically low-contrast lesions, for other observers (for example channelised Hotelling) and for search tasks is untested.

Tenth, the behaviour under distribution shift depends on the reconstruction condition, and the part that is common to all conditions is accuracy: on non-anatomical phantoms the learned prior is below TV in PSNR and SSIM in every condition tested, and its detection advantage is retained in one of six conditions. The number of false lesion-like blobs relative to TV changes sign with the number of angles, because the TV baseline's own count varies about 26-fold across conditions; relative to FBP-Hann the prior adds a few false blobs per phantom at quarter dose (+1.0 to +4.0 at 1× lesion strength) and removes many at full dose. We therefore do not claim that the prior hallucinates in general or that it is safe: the evidence is that it does not create many lesion-like structures on real slices in distribution, and that under this shift it behaves as a moderately worse reconstructor whose false-blob count is neither consistently higher nor lower than that of its competitors.

Answers to the guiding questions. (1) Does the learned prior keep its advantage over TV when the image distribution changes? Its detection advantage, the only one it had on real slices, is kept in one of six shift conditions, significantly smaller than on CT in one ($K = 120$, quarter dose) and not significantly different from CT in the three other conditions where this was tested ($K = 60$ was not tested); its PSNR and SSIM deficit against TV grows. (2) Can we find cases where the

method creates convincing but incorrect structures? Yes, at quarter dose on phantoms (a few blobs per phantom more than FBP-Hann, and in the $K = 120$ conditions more than TV); not on real slices, where it creates fewer such blobs than FBP-Hann in all four conditions examined and at most 1.4 per slice at lesion strength. The structures it creates there are of the same kind that the data produce in FBP and TV. (3) Do the conclusions hold over noise levels and reconstruction conditions? The sign of the detection advantage over TV does (nine of nine, significant in six); the conclusions on PSNR, SSIM and false blobs do not: their signs change with dose and angle count without a monotone trend.

6 Conclusion

We presented a controlled, small-scale comparison of classical and learned regularization for sparse-view CT reconstruction under noise, angular undersampling, forward-model mismatch, distribution shift and a lesion-detection task. The learned Plug-and-Play prior outperforms TV by up to $\approx 4\,\mathrm{dB}$ at moderate-to-high noise and for 5–90 angles, but the advantage is conditional: the near noise-free ranking depends on the data model, the margin narrows under forward-model mismatch, and it reverses under distribution shift. The reversal is caused by a mismatched prior rather than by the PnP scheme: a prior trained on the shifted distribution beats TV on it, and the mismatched prior fills the weakly determined subspace with up to three times the true energy. In a task-based evaluation, the PSNR-tuned PnP reconstruction attenuates small lesions in distribution and invents lesion-like blobs under mismatch; re-tuning recovers TV-level detection only by giving up the PSNR advantage or by increasing false structure, except for a matched prior. The iteration used here lacks a stable, accurate fixed point, so early stopping acts as a regularizer. On clinical-size CT from LoDoPaB, with patient-disjoint splits, nine dose and angle conditions and exact Poisson dose thinning, the learned prior does not improve on a well-tuned TV in PSNR or SSIM, but removes the detection deficit that PSNR-tuned TV has relative to filtered back-projection (AUC higher than TV in all nine conditions, equal to FBP-Hann within noise in seven). Under a shift to phantoms its accuracy deficit against TV is consistent and its detection advantage is mostly not demonstrable, and the number of invented lesion-like blobs is not systematically different from that of TV or FBP-Hann beyond a few per phantom at quarter dose. Future work should use all 60 test patients with several noise realisations, a finer TV and Hann grid, several denoiser seeds and the full training set, clinically calibrated lesion contrasts with model observers, and provably convergent PnP variants.

Limitations

The study is intentionally small (28×28 MNIST-derived images) so that a thorough experimental grid is possible; trends may not transfer to natural-image or clinical CT resolution and geometry. Experiments E1–E5 use data generated with the reconstruction operator (an inverse crime); the crime-free experiments E6–E10 use bicubic finer-grid data, and the sensitivity band of Table 4 shows that the near noise-free ranking depends on this choice. The convergence study (E5) and the forward-model mismatch study (E3) were not repeated on finer-grid data. Seed variation is estimated from three denoisers per configuration, which gives only a crude standard deviation, and the Shepp–Logan test set is a single image with 20 noise draws, so its intervals reflect noise rather than image variation. The error-decomposition statistics cannot separate prior-driven structure from amplified noise except in comparisons with identical noise and settings, and the observability analysis of lesions is confounded with lesion size and location. The lesion study uses bright additive

Gaussian blobs, a location-known observer and a false-lesion threshold that is a modelling choice, so false-lesion rates are meaningful for comparing methods but not as absolute rates; the operating-point study uses one denoiser per prior, two lesion types and a coarse grid. Only two training distributions were compared, so the explanation of the asymmetric transfer between them remains a hypothesis. The PnP scheme has no convergence guarantee, and only moderate forward-model mismatches (up to 4°) were evaluated. The Poisson study uses a single attenuation scale ($\mu = 2$), which keeps the signal dependence of the noise mild, a single angle count, and one MNIST-trained denoiser in the dose sweep, so the shift results of E12 carry no seed uncertainty; its Shepp–Logan result is one image and one noise draw per dose. The lesion study under Poisson noise inherits the limitations of E9, its TV baseline is a single run, and its observability analysis is exploratory and image-set dependent. The matrix-free operator was validated only against the dense operator at 28×28, not at clinical size, where the cost of the matrix-free TV solver will have to be reduced. The CT study uses 40 test patients (one slice each) and one noise realisation per condition, so a single cell has an AUC uncertainty of about ± 0.05 and the apparent pattern across the nine conditions rests on sign consistency; the conditions share the same patients and are not independent. No correction for multiplicity was applied, the analysis was developed in steps after seeing results (Section 3.16), and the shift extension was chosen after the $K = 60$ results. One denoiser architecture and one training run were used, with 8 960 of the 35 820 training slices, so the comparison does not separate the prior from this particular network. TV is tuned on a grid with steps of a factor three, TV-long is sometimes worse than TV, and PSNR was the tuning criterion for every method, which favours accuracy over detectability. Two optima of PnP-TVinit lie on the boundary of the grid. The simulator was checked against real data on 8 slices per condition and agrees within 0.15 dB in TV PSNR, but the real data have more variance than the simulation at high counts. The lesions are additive Gaussian blobs of 120–240 HU under an assumed mapping from $\mu/\mu_{\max}$ to HU, the observer is a location-known statistic, false blobs are defined by this statistic and a threshold, and no reader study was done. The phantoms are piecewise constant, which favours TV; the shift study covers a single kind of shift and no change of scanner geometry or dose model. The $K = 60$ shift analysis used four lesion types and the other conditions two; PSNR on the phantoms at $K = 60$ is reported without an interval. The DnCNN has no convergence guarantee and the number of iterations acts as a regulariser.

Reproducibility

Code, trained models, all raw result files and figures are available at https://github.com/AboOraig/learned-regularization. The experiments of this version are run, in order, by `run_nullspace.py` (E6, E7), `run_spectral.py` (iteration and spectral analysis), `run_step4.py` and `summarize_step4.py` (E8), `run_lesions.py` and `summarize_step5.py` (E9), and `run_oppoint.py` (E10); `run_stage2.py` (E11, E12) and `run_lesions_poisson.py` with `summarize_step5.py` (E13); `run_stage3_step1.py` (data, patient split, geometry, E14), `run_stage3_step2.py` (noise model, FBP and TV baselines, extraction of training slices), `train_denoiser_ct.py` (denoiser), `run_stage3_step3.py` and `run_stage3_step3b.py` (E15; the first is superseded), `run_stage3_step4.py`, `run_stage3_step4b.py` and `run_stage3_step4c_compare.py` (E17b, E18, E19 at $K = 60$; use `--out results/stage3/s4c --n_val 20 --auto_extend 2` for the final version), `run_stage3_step5.py` and `run_stage3_step5_summary.py` (E16–E18), and `run_stage3_step5_shift.py` with `run_stage3_step5_shift_summary.py` (E19 at $K = 30, 120$); the unit tests (68 in total) run with `python -m pytest tests`.

References

[1] Vegard Antun, Francesco Renna, Clarice Poon, Ben Adcock, and Anders C. Hansen. On instabilities of deep learning in image reconstruction and the potential costs of AI. *Proceedings of the National Academy of Sciences*, 117(48), 2020.

[2] Harrison H. Barrett and Kyle J. Myers. *Foundations of Image Science*. Wiley, 2004.

[3] Somesh Bhadra, V. A. Kelkar, Frank J. Brooks, and Mark A. Anastasio. On hallucinations in tomographic image reconstruction. *IEEE Transactions on Medical Imaging*, 40(11), 2021.

[4] A. Chambolle and T. Pock. A first-order primal-dual algorithm for convex problems with applications to imaging. *Journal of Mathematical Imaging and Vision*, 40(1):120–145, 2011.

[5] Stanley H. Chan, X. Wang, and O. A. Elgendy. Plug-and-play ADMM for image restoration: Fixed-point convergence and applications. *IEEE Transactions on Computational Imaging*, 3 (1), 2017.

[6] M. Z. Darestani, A. S. Chaudhari, and Reinhard Heckel. Measuring robustness in deep learning based compressive sensing. In *International Conference on Machine Learning (ICML)*, 2021.

[7] Nathan Halko, Per-Gunnar Martinsson, and Joel A. Tropp. Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions. *SIAM Review*, 53(2):217–288, 2011.

[8] Jari Kaipio and Erkki Somersalo. *Statistical and Computational Inverse Problems*. Springer, 2005.

[9] Johannes Leuschner, Michael Schmidt, Daniel Otero Baguer, and Peter Maass. LoDoPaB-CT, a benchmark dataset for low-dose computed tomography reconstruction. *Scientific Data*, 8: 109, 2021.

[10] Leonid I. Rudin, Stanley Osher, and Emad Fatemi. Nonlinear total variation based noise removal algorithms. *Physica D: Nonlinear Phenomena*, 60(1–4), 1992.

[11] Ernest K. Ryu, J. Liu, S. Wang, X. Chen, Z. Wang, and Wotao Yin. Plug-and-play methods provably converge with properly trained denoisers. In *International Conference on Machine Learning (ICML)*, 2019.

[12] Singanallur V. Venkatakrishnan, Charles A. Bouman, and Brendt Wohlberg. Plug-and-play priors for model based reconstruction. In *IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, 2013.

[13] Armand Wirgin. The inverse crime, 2004.

[14] Kai Zhang, Wangmeng Zuo, Yunjin Chen, Deyu Meng, and Lei Zhang. Beyond a gaussian denoiser: Residual learning of deep CNN for image denoising. *IEEE Transactions on Image Processing*, 26(7), 2017.